



Collective-level propagation analysis via LLM-enhanced hypergraph transformer for fake news detection

Siqi Wei^a, Bin Wu^a,^{*}, Tao Zhou^b, Xuanning Liu^a, Jiachen Tan^a, Di Liu^a, Guangyao Su^b

^a School of Computer Science, Beijing University of Posts and Telecommunications, Beijing, China

^b China Unicom Software Research Institute, Beijing, China

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ABSTRACT

Fake news detection is crucial for safeguarding information security and maintaining social stability. Existing studies use graph neural networks to capture news propagation features to boost fake news detection. However, constrained by the pairwise nature of conventional graphs, they decompose propagation into a set of individual-level behaviors. In practice, news diffusion is a collective propagation process rather than a simple aggregation of individual behaviors. Real and fake news exhibit distinct collective propagation behavior patterns, which can provide valuable cues for fake news detection. To exploit such collective-level characteristics, we propose LeHT, an LLM-enhanced hypergraph transformer framework for fake news detection via collective-level propagation analysis. LeHT leverages a large language model as a context provider to generate context-enhanced semantic representations, alleviating semantic sparsity in short news posts and enabling better understanding of semantics that may trigger collective reactions. Meanwhile, we construct propagation hypergraphs to explicitly model collective propagation behaviors, where each news item is represented as a hyperedge connecting all participating users. Based on the propagation hypergraph and the context-enhanced semantic representations, we design a hypergraph transformer network with a three-level attention mechanism to capture complex higher-order correlations in collective propagation behaviors and learn discriminative collective propagation behavior pattern representations for fake news detection. Experiments on three public datasets — Twitter (1152 news items), PHEME (1705 news items), and DRWeibo (6037 news items) — demonstrate that LeHT achieves ACC/F1 scores of 0.923/0.921, 0.926/0.924, and 0.916/0.913, respectively.

1. Introduction

The rapid development of social media has profoundly transformed the production and dissemination of information (Bondielli & Marcelloni, 2019). While social platforms facilitate instant information sharing and broad public access, they also provide a convenient channel for the creation and rapid diffusion of fake news. The widespread dissemination of fake news not only undermines information credibility but may also trigger social unrest (Hu et al., 2022). Therefore, fake news detection has become a critical task for safeguarding information security and maintaining social stability (Zhou & Zafarani, 2020).

* Corresponding author.

E-mail addresses: weisqi@bupt.edu.cn (S. Wei), wubin@bupt.edu.cn (B. Wu), jc-zhout2@chinaunicom.cn (T. Zhou), liuxuanning@bupt.edu.cn (X. Liu), tanjiaichen@bupt.edu.cn (J. Tan), liudi@bupt.edu.cn (D. Liu), sugy@chinaunicom.cn (G. Su).

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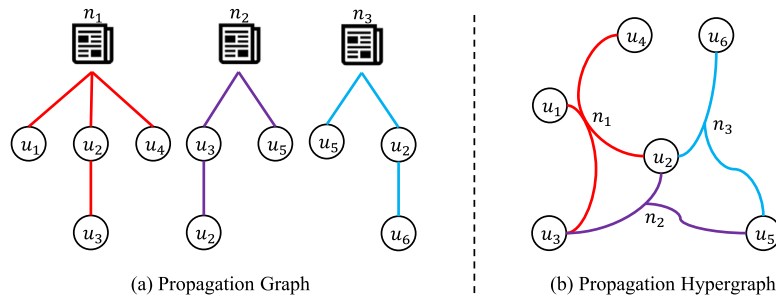


Fig. 1. Comparison between propagation graph and propagation hypergraph. Graph structures model only pairwise correlations and primarily capture individual-level propagation behaviors, whereas hypergraphs allow a single hyperedge to connect multiple users, enabling explicit modeling of collective propagation behaviors.

Traditional fake news detection methods (Chen et al., 2025; Cheng et al., 2020; Hande et al., 2021; Hu et al., 2024; Ma et al., 2016; Qin & Zhang, 2024; Raza et al., 2025; Schütz et al., 2021; Yu et al., 2017; Zhang et al., 2025) mainly rely on content analysis, extracting linguistic and semantic features from news text. However, fake news is often deliberately crafted to mimic real news, making it difficult to distinguish misinformation from truthful content based solely on textual cues. To address this limitation, a growing body of work incorporates propagation signals as complementary evidence. Propagation-based methods (Bian et al., 2020; Cui & Jia, 2024, 2025; Guo et al., 2025a; Huang et al., 2025; L. Li et al., 2025; Li et al., 2020; G. Li et al., 2025; Song et al., 2022; Wu et al., 2024) typically model news diffusion processes with graph structures and employ graph neural networks (GNNs) to capture structural and temporal characteristics of news spread, thereby achieving clear improvements over purely content-based methods.

Despite these advances, existing propagation-based approaches remain fundamentally limited by the pairwise nature of conventional graph structures. As illustrated in Fig. 1(a), each edge in a propagation graph usually represents one user reposting a news item under the influence of a single predecessor, thereby modeling news diffusion as an accumulation of individual-level pairwise propagation actions. However, social-psychological mechanisms such as bandwagon (Molina et al., 2023) and conformity (Ma et al., 2025) effects suggest that users are often influenced by the already formed propagation collective rather than merely by a single predecessor. Studies on information diffusion and social psychology (Lu et al., 2020; Ma et al., 2025; Molina et al., 2023) suggest that collective propagation behavior refers to the phenomenon that, under the joint influence of news semantics and visible collective signals released by the already formed propagation collective, multiple users make propagation decisions in an interdependent manner. Since such group-level influence is inherently higher-order in nature, it cannot be adequately captured by simply accumulating pairwise reposting relations.

Such higher-order collective influence leaves observable collective propagation behavior patterns. Unlike individual-level propagation patterns, which mainly describe local reposting behaviors or pairwise influence, collective propagation behavior patterns are macro-level regularities emerging from the interdependent decisions of a group of users. Social-psychological mechanisms such as the bandwagon effect (Molina et al., 2023) and the conformity effect (Ma et al., 2025) provide an important explanation for the transition from interdependent behavior to observable patterns. Once visible collective signals become salient, subsequent users may become more likely to join the diffusion process by following the already formed collective behavior. As a result, the diffusion process may leave recognizable collective-level traces, including temporal signals, structural signals, and comment-semantic signals. Temporal signals describe how collective participation unfolds over time; structural signals reflect how users become jointly involved in the same diffusion process; and comment-semantic signals characterize the collective semantic responses formed during diffusion. These observable traces jointly constitute collective propagation behavior patterns and cannot be adequately captured by individual-level propagation modeling alone.

Empirical studies (Vosoughi et al., 2018) show that real and fake news differ substantially in collective propagation behavior patterns. From a collective-level perspective, in the early stage of diffusion, fake news is more likely to form semantically aligned, tightly coordinated, and burst-like propagation collectives, whereas real news tends to develop through a more gradual, organic, and semantically heterogeneous collective process from the outset. Specifically, fake news is often crafted with sensational, emotional, or polarizing semantics, which can quickly attract an initial group of users with aligned motivations to actively disseminate the content. In some cases, such diffusion may also be reinforced by coordinated actors or social bots, allowing the news to rapidly accumulate strong visible collective signals. Under such conditions, later users are more likely to be drawn into the diffusion process through bandwagon and conformity effects, making the early-stage propagation of fake news more collective, more synchronized, and more amplifiable. In contrast, real news is more often reported in a relatively objective and information-oriented manner, with weaker sensational or polarizing semantics. Consequently, it is less likely to induce immediately aligned collective reactions and is more often propagated by users with heterogeneous motivations, such as information sharing, discussion, verification, or debate. Therefore, the collective formation of real news tends to be more gradual, more organic, less coordinated, and more semantically diverse from the beginning. Meanwhile, fake news may show another distinctive characteristic in the later stage: as debunking information, skepticism, and corrective discussions emerge, its collective semantic responses can become more heterogeneous or

even conflicting. Therefore, fake and real news differ not only in content, but also in the formation logic and temporal evolution of their collective propagation behavior patterns.

Motivated by these observations, we advocate collective-level propagation analysis for fake news detection. The core idea is to move beyond modeling isolated user interactions and instead capture the higher-order propagation patterns formed by groups of users. However, effective collective-level propagation analysis faces two major challenges.

- **Challenge 1: Semantic understanding under short-text sparsity.** Mining collective-level propagation patterns requires understanding the news semantics that may trigger collective reactions. In practice, fake news often contains sensational, emotional, or polarizing semantics that can rapidly attract aligned disseminators and initiate early collective amplification. However, social media news posts are typically short and context-poor, leaving many implicit details (e.g., event background and entity attributes) unstated. This semantic sparsity makes it difficult to accurately interpret post content and identify the semantic triggers of collective propagation.
- **Challenge 2: Capturing higher-order correlations in collective propagation.** Collective propagation is characterized by higher-order dependencies beyond pairwise interactions. For example, users within a propagation collective may co-spread the same news item, while later participants may join the diffusion process under the joint influence of the already formed collective. In conventional graph structures, such higher-order collective dependencies are decomposed into multiple pairwise edges, thereby diluting or even losing the critical collective semantics. This structural limitation poses a challenge for mining collective-level propagation pattern features.

To address these challenges, we propose **LeHT**, an LLM-enhanced **Hypergraph Transformer** framework for fake news detection via collective-level propagation analysis. **To address Challenge 1**, we leverage large language models (LLMs) as context providers to alleviate semantic sparsity in short news posts. LLMs are enhanced versions of pre-trained language models, containing rich world knowledge and showing strong capability across a wide range of natural language processing tasks (J. Li et al., 2025). LeHT prompts an LLM to generate multi-granularity context candidates, including coarse-grained event background and fine-grained entity descriptions, and then selectively integrates them with the original news text to alleviate semantic sparsity and enrich news semantics. The resulting context-enhanced representations help the model better understand the semantics that may trigger collective responses, thereby facilitating the discovery of collective-level propagation patterns. For deployment efficiency, we further derive a lightweight distilled variant that approximates the teacher-induced semantic enhancement effect without requiring LLM queries at inference time. **To address Challenge 2**, we introduce hypergraphs to explicitly model collective propagation behaviors in social networks. A hypergraph is an extension of a graph consisting of node sets and hyperedge sets (Lee et al., 2025). As shown in Fig. 1, unlike conventional pairwise graphs, each hyperedge in a hypergraph can connect any number of nodes, making it naturally suitable for modeling higher-order correlations. In this paper, for each target news item, we construct a sample-specific cumulative propagation hypergraph using only historically observable propagation records up to its prediction time, where users are treated as nodes and each news item is represented as a hyperedge connecting the users who participate in propagating it. This construction preserves crucial collective-level features in news diffusion. Importantly, it models collective participation in the same diffusion event rather than stance homogeneity. Building on the selectively enhanced semantic representations and the constructed propagation hypergraph, we design a hypergraph transformer network (HTN) to learn collective propagation behavior pattern representations for fake news detection. Specifically, the HTN incorporates a three-level hypergraph attention mechanism to capture higher-order collective dependencies: (i) hyperedge-to-node attention aggregates the features of propagated news to obtain a global credibility representation of each user; (ii) behavior-aware node-to-node attention introduces temporal, structural, and comment-semantic signals to refine node representations and produce news-specific user representations; and (iii) node-to-hyperedge attention aggregates user features within each collective to produce collective propagation pattern representations.

Finally, we conduct extensive experiments on three public fake news detection datasets, including Twitter (Ma et al., 2017), PHEME (Kochkina et al., 2018), and DRWeibo (Cui & Jia, 2024). Experimental results show that LeHT outperforms state-of-the-art fake news detection methods. Moreover, LeHT can detect fake news before it achieves widespread diffusion, which is valuable for mitigating the harmful influence of misinformation.

The main contributions of this work are summarized as follows:

- **Problem-solving perspective.** We focus on collective-level propagation analysis, which aims to identify differences in collective propagation behavior patterns between real news and fake news, providing a novel perspective for fake news detection.
- **Methodology.** We propose an LLM-enhanced hypergraph transformer framework (LeHT) that detects fake news through collective-level propagation analysis. The framework leverages both the rich world knowledge of LLMs and the strong higher-order correlation modeling capability of hypergraphs.
- **Empirical validation.** Extensive experiments on three public datasets demonstrate that LeHT achieves superior performance over state-of-the-art comparative methods, confirming the effectiveness of collective-level propagation analysis for fake news detection.

2. Related work

The fake news detection task aims at automatically distinguishing the authenticity of news based on its features (Shu et al., 2017). Based on the selected features, existing fake news detection works can be broadly categorized into the following two groups: content-based and propagation-based approaches.

2.1. Content-based methods

Content-based methods extract semantic features from news text content to detect fake news. Existing approaches can be broadly categorized into end-to-end models, pre-trained language model (PLM)-based models, and large language model (LLM)-based models.

End-to-end methods train neural networks from scratch using a single fake news detection dataset. For example, Ma et al. (2016) employ recurrent neural networks (RNNs) to encode textual representations before feeding them into a classifier. Yu et al. (2017) adopt convolutional neural networks (CNNs) to learn news text features. Cheng et al. (2020) introduce a multi-task autoencoder framework to enhance textual understanding. However, due to the limited size of the single fake news dataset, end-to-end models may struggle to learn semantic representations well, which severely constrains their generalization ability.

Pre-trained language models (PLMs) acquire strong semantic understanding capabilities through self-supervised learning on large-scale corpora. With the success of pre-trained language models, PLM-based methods fine-tune pre-trained models for downstream fake news detection tasks. Hande et al. (2021) fine-tune BERT (Devlin et al., 2019) and DeBERTa (He et al., 2021) on multiple FND benchmarks, while Schütz et al. (2021) fine-tune XLNet (Yang et al., 2019) for the same purpose. More recently, Qin and Zhang (2024) propose an adversarial fine-tuning strategy to improve the generalization of BERT-based detectors.

Large language models, as upgraded versions of pre-trained language models, contain more parameters and larger training corpora. LLMs have demonstrated strong capabilities on numerous NLP tasks (J. Li et al., 2025). Consequently, recent studies have explored leveraging LLMs for fake news detection. Based on the role of LLMs in the detection pipeline, existing methods can be grouped into LLM-as-a-classifier and LLM-as-an-advisor paradigms. In the first paradigm, LLMs directly perform fake news classification through prompt engineering. For instance, Raza et al. (2025) evaluate the zero-shot FND performance of mainstream LLMs such as GPT-3.5 and GPT-4. In the second paradigm, the LLM acts as an advisor to generate rationales that aid in decision-making, and an additional small model is trained to incorporate these rationales for fake news detection. Hu et al. (2024) instruct LLMs to generate rationales from the perspectives of common sense and textual descriptions, and design an adaptive rationale guidance network to selectively acquire insights on news analysis from the LLMs' rationales. Chen et al. (2025) also prompt LLMs to generate rationales for detecting fake news, then propose a graph-based model to fuse LLM-generated rationales with news content for enhanced reasoning. Zhang et al. (2025) generate both positive and negative rationales and introduce a semantic consistency objective that encourages news representations to align with positive rationales while diverging from negative ones.

Although LLM-based approaches achieve some success, fake news is often deliberately crafted to mimic real news, making it insufficient to rely solely on content. Furthermore, social media news posts are typically short and lack essential contextual information, which poses significant challenges for gaining a deep understanding of news semantics. To address these limitations, we propose a collective-level propagation analysis framework, LeHT, which leverages macro-level collective behavior patterns to provide complementary evidence beyond textual content. In addition, we introduce an LLM-driven context enhancement module that prompts an LLM to generate rich contextual information for short news posts, thereby mitigating semantic sparsity and enabling more reliable detection.

2.2. Propagation-based methods

Propagation-based approaches utilize graph structure to model the news propagation process and employ graph neural networks (GNNs) to learn propagation representations for fake news detection. Depending on the adopted graph structure, existing methods can be grouped into simple graph-based, dynamic graph-based, and hypergraph-based approaches.

A simple graph consists of nodes and pairwise edges, where nodes represent users and edges represent retweet relations. Simple graph-based methods construct static news propagation graphs and design various GNN architectures to extract structural clues for fake news detection. Li et al. (2020) employ GraphSAGE to learn structural representations from propagation networks. Bian et al. (2020) design a top-down and bottom-up graph convolutional network to capture both propagation and diffusion signals. Wu et al. (2024) use a bidirectional GCN combined with CNN-based fusion to extract multi-perspective propagation features. L. Li et al. (2025) adopt GAT to encode news propagation. G. Li et al. (2025) propose an emotion-aware structural enhancement graph autoencoder to obtain emotion-enriched propagation representations to distinguish real news and fake news. Cui and Jia (2024) propose an adaptive graph contrastive learning method to help models acquire more discriminative news representations. Zheng et al. (2024) proposes a hierarchical prototype contrastive learning mechanism to pull news closer to its corresponding semantic prototypes while pushing it farther from other semantic prototypes, thereby enhancing the semantic representation capabilities of graph neural networks. Cui and Jia (2025) incorporate supervised contrastive learning and unsupervised contrastive learning auxiliary tasks into fake news detection, thereby enhancing model performance. Although these methods can effectively capture structural signals in the news propagation process, they neglect temporal signals.

To incorporate temporal information in the news propagation process, dynamic graph-based approaches model the evolution of propagation networks over time. Song et al. (2022) represent each news item as a sequence of graph snapshots and learn temporal propagation dynamics through dynamic GNNs. Huang et al. (2025) construct two dynamic graphs (including an information propagation graph and a user interaction graph) and propose a multi-view spatiotemporal GNN to jointly capture structural and temporal evolution. Guo et al. (2025b) proposes an adaptive sliding window mechanism to partition nodes of a rumor propagation graph into multiple disjoint windows for modeling the dynamic news propagation process, and introduces a memory-enhanced attention mechanism to capture temporal evolution features. Guo et al. (2025a) utilize edge timestamps to build edge-weighted dynamic propagation graphs and develop a fine-grained dynamic GNN to learn detailed temporal interaction patterns. Despite

achieving some results, nodes in dynamic graphs remain pairwise connected. Consequently, these dynamic graph-based methods struggle to model higher-order correlations inherent in the collective propagation behavior.

Some studies incorporate hypergraphs into fake news detection tasks. Dong et al. (2022) introduce a sentiment-aware hypergraph attention network to capture higher-order interactions among comments sharing the same sentiment polarity. Li et al. (2023) develop a conversation-branch-tweet hypergraph convolutional network to model higher-order correlations among comments within the same conversation branch. Su et al. (2025) leverage the hypergraph convolution network to encode higher-order user-news interactions, where a user's historically shared news jointly reflects the user's credibility, and then combine the learned credibility signals with content features for detection. Similarly, HG-SL (Sun et al., 2023) employs a hypergraph neural network to learn global user credibility and further introduces temporal and structural self-attention modules to capture the temporal and structural features of propagation.

Despite achieving some results, existing hypergraph-based approaches typically focus on higher-order correlations among specific interaction units in the news propagation process (e.g. sentiment-consistent comments, comments within the same conversation branch, or posts associated with the same user). They still struggle to explicitly model news propagation at the macro collective level, making it challenging to analyze the differences in collective propagation behavior patterns between real news and fake news. Moreover, many of the above methods rely on short, context-poor news posts and thus overlook the semantic sparsity issue, which can hinder deep semantic understanding and, consequently, the modeling of collective propagation patterns.

In contrast, our approach leverages hypergraphs to explicitly model collective propagation behaviors in the social network. We further introduce an LLM-driven context enhancement module to enrich short posts with multi-granularity contextual information. By combining the powerful higher-order correlation modeling capabilities of hypergraphs with the rich world knowledge of large language models (LLMs), our proposed LLM-enhanced hypergraph transformer, LeHT, can more effectively capture collective propagation behavior pattern features and distinguish real news from fake news.

3. Research objective

In this section, we formally introduce our research objective: collective-level propagation analysis. Before proceeding, we first introduce some relevant notions and definitions.

Notations. Let $\mathcal{N} = \{n_1, n_2, \dots, n_{|\mathcal{N}|}\}$ denote the set of news items, where $|\mathcal{N}|$ is the number of news items. Each news item n_j is associated with an authenticity label $y_j \in \{0, 1\}$, indicating whether the news is real or fake. Let $\mathcal{U} = \{u_1, u_2, \dots, u_{|\mathcal{U}|}\}$ denote the set of users in the online social network, where $|\mathcal{U}|$ is the number of users.

In social platforms, users can post or repost news items, forming observable propagation behaviors. We represent each user-news interaction as a propagation record $r_{i,j} = \{u_i, n_j, t_{i,j}, c_{i,j}, d_{i,j}, b_{i,j}\}$, which indicates that user u_i participates in the propagation of news item n_j . Specifically, $t_{i,j}$ denotes the timestamp when user u_i propagates n_j , $c_{i,j}$ denotes the corresponding user comment, $d_{i,j}$ denotes the propagation depth of user u_i in the diffusion of n_j , and $b_{i,j}$ denotes the number of subsequent reposts triggered by user u_i . These propagation records jointly characterize the temporal, structural, and semantic aspects of news propagation. In addition, let r_j^{pub} denote the publication time of news item n_j , and let T_j denote the prediction time when the authenticity of n_j is to be determined.

For a target news item n_j , we further use $\mathcal{U}_j^{\leq T_j}$ to denote the set of users who have participated in propagating n_j by time T_j , and use $\mathcal{N}_i^{\leq T_j}$ to denote the set of news items that user u_i has propagated by time T_j .

Definition. Hypergraph. A hypergraph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_i\}_{i=1}^N$ denotes the set of vertices and $\mathcal{E} = \{e_j\}_{j=1}^M$ denotes the set of hyperedges. Unlike pairwise connections in a conventional graph, a hypergraph allows one hyperedge to connect any number of vertices. A hypergraph can be represented by an incidence matrix $\mathbf{A} \in \mathbb{R}^{N \times M}$, where $A_{i,j} = 1$ if hyperedge e_j contains vertex v_i , and $A_{i,j} = 0$ otherwise.

Collective-level propagation analysis

Our research objective is to detect fake news through collective-level propagation analysis. Collective-level propagation analysis aims to capture collective propagation behavior patterns and distinguish real news from fake news according to the differences in these patterns. Unlike existing studies that mainly model individual-level propagation behaviors, collective-level propagation analysis focuses on the macro-level patterns jointly formed by the users participating in the same news diffusion event.

Specifically, for each target news item n_j , we construct a sample-specific cumulative propagation hypergraph using only historically observable propagation records up to its prediction time T_j . In this hypergraph, users serve as nodes, while each news item serves as a hyperedge connecting the users who participate in propagating it. This formulation models collective participation in the same diffusion event rather than stance homogeneity, and preserves higher-order co-propagation relationships among users. Since the hypergraph is constructed cumulatively up to T_j , the authenticity prediction of n_j depends only on historically available propagation evidence and does not access any future news item or future propagation record.

Based on the constructed sample-specific cumulative propagation hypergraph, collective-level propagation analysis reformulates fake news detection as the classification of the target hyperedge corresponding to n_j . The authenticity of the target news is inferred from the collective propagation behavior pattern formed by the user group participating in its diffusion.

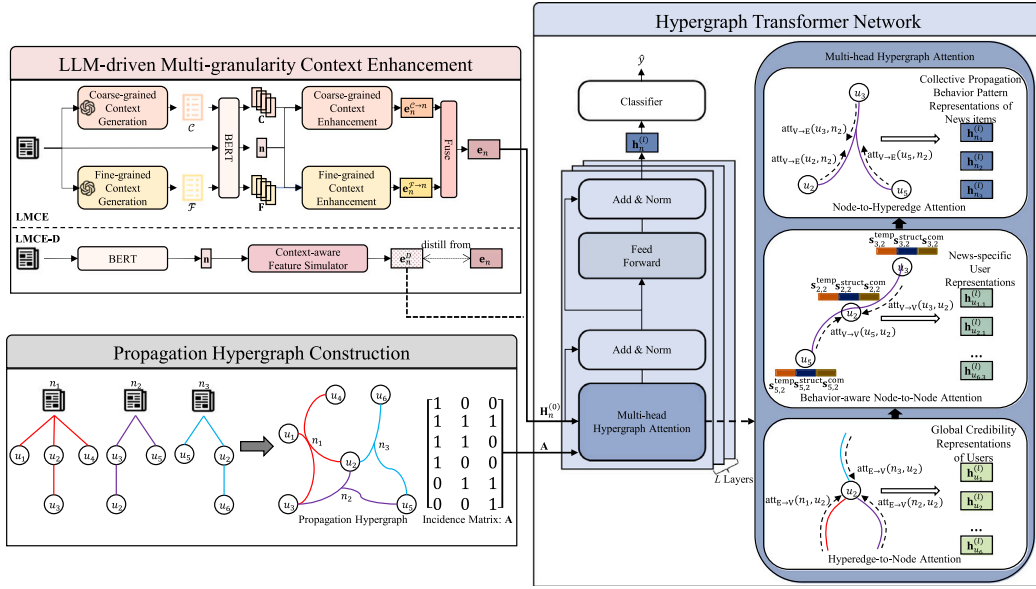


Fig. 2. The architecture of the LeHT framework.

4. Methodology

4.1. Overview

In this paper, we propose LeHT, an LLM-enhanced hypergraph transformer framework for fake news detection via collective-level propagation analysis. The overall architecture of LeHT is illustrated in Fig. 2. LeHT comprises three main modules. First, the LLM-driven multi-granularity context enhancement (LMCE) module prompts an LLM to generate multi-granularity contextual candidates and selectively integrates them with the original news text to produce a context-enhanced semantic representation. Second, the sample-specific cumulative propagation hypergraph construction module builds a propagation hypergraph for each target news item using only historically observable propagation records up to its prediction time, so as to explicitly model collective propagation behaviors. Third, based on the context-enhanced semantic representations and the constructed sample-specific cumulative propagation hypergraph, the hypergraph transformer network (HTN) captures higher-order associations in collective propagation behaviors through the proposed three-level hypergraph attention mechanism and generates collective propagation behavior pattern representations for fake news detection.

4.2. LLM-driven multi-granularity context enhancement

News posts contain semantics that may trigger collective responses. Therefore, mining collective propagation behavior patterns requires deep understanding of news semantics. However, news posts on social media platforms are typically brief and contextually sparse, which severely limits comprehensive semantic understanding. To alleviate this issue, we propose an LLM-driven multi-granularity context enhancement (LMCE) module. LMCE prompts an LLM to generate multi-granularity contextual candidates for the target news and then selectively integrates them with the original news representation to obtain a context-enhanced semantic representation. For cost-sensitive deployment scenarios, we further introduce an LLM-free distilled variant, termed LMCE-D.

4.2.1. Multi-granularity context generation

To enrich the semantics of short and context-sparse news posts for downstream collective-level propagation analysis, we design a set of reasoning-guided prompts to guide the large language model (LLM) to generate multi-granularity contextual candidates. This multi-granularity design is intended to enrich the semantics of news from various perspectives, thereby enhancing the robustness of the model. Importantly, the LLM in our framework is not used as a classifier or a decision-maker. Instead, it serves only as a context provider that supplies auxiliary semantics for the target post.

Although LLMs perform well on many NLP tasks, they are still at risk of generating hallucinations. When designing the prompts, we explicitly take the risk of LLM hallucination into account. Specifically, our prompt design follows five principles: (1) objective generation, which prevents the LLM from treating the claim in the post as true by default and prohibits fact-checking or veracity

judgment, thereby encouraging only objective and non-opinionated contextual outputs; (2) relevance-oriented generation, which constrains the LLM to produce only event- or entity-level information directly relevant to the target post, preventing overly expansive or tangential generation; (3) high-credibility generation, which requires the LLM to rely only on highly reliable and widely established knowledge; (4) conservative generation, which requires the LLM to use cautious and generic wording when event details or entity identities are ambiguous; and (5) structured generation, which constrains the output to a predefined JSON format for subsequent processing.

Formally, a prompt is defined as a triple $p = \langle \tau, n_j, \varepsilon \rangle$, where τ denotes the task instruction, n_j is the input news text of the j th news, and ε specifies the output format. According to the granularity of contextual cues, we design two prompt templates to generate coarse-grained and fine-grained contextual candidates, respectively.

Coarse-grained context generation. Coarse-grained context aims to provide background information for the core event described in the news post. Social media news typically reports or comments on some events, so understanding the event background is crucial for correctly interpreting the news semantics. However, due to length constraints, such background information is often omitted, leading to incomplete semantic understanding. To mitigate this limitation, we design a reasoning-guided prompt that instructs the LLM to (i) identify the core event mentioned in the news post and (ii) generate concise background information that facilitates understanding of the event.

To mitigate hallucinations, the coarse-grained prompt explicitly instructs the LLM not to treat claims in posts as true by default, not to perform fact-checking or authenticity judgments, and to generate only objective contextual information. Furthermore, the prompt explicitly instructs the LLM not to introduce information unrelated to the news post. It further requires the model to generate only highly relevant and reliable background information regarding the described event. Additionally, it specifies that conservative and general language should be used when the event description is ambiguous. In this way, the LLM is encouraged to generate stable, event-level contextual candidates rather than speculative or over-elaborated narratives. The prompt template is shown below.

Prompt for Coarse-grained Context Generation

Task Description

You are a senior editor in journalism. Given a news post from social media, your task is to: (1) identify the core event described in the news post, and (2) provide concise background information relevant to understanding this event. Follow the two-step process internally, but output only the final result.

Important requirements:

- Do not assume that the claim in the news post is true, and do not perform fact-checking or veracity judgment. Produce only objective background information about the event. Do not include subjective opinions or personal speculation.
- Only provide background information that is highly relevant to understanding the mentioned event.
- Provide only highly reliable and widely known background information about the event.
- Do not add specific details that are not widely confirmed background knowledge. If some event details are ambiguous, use conservative and general wording rather than completing missing details.

Input Data

Given news post: $[n_j]$

Output Format

Return the result in the following JSON format: $\{ "C_j": ["c_{j,1}", "c_{j,2}", \dots] \}$

We then query the LLM with the coarse-grained prompt template to obtain a list of coarse-grained contextual candidates:

$$C_j = \text{LLM}(p_{\text{CCG}}(\tau_{\text{CCG}}, n_j, \varepsilon_{\text{CCG}})), \quad (1)$$

where $C_j = [c_{j,1}, c_{j,2}, \dots, c_{j,|C_j|}]$ denotes the coarse-grained contextual candidates generated for the j th news post, p_{CCG} denotes the coarse-grained prompt template, and $|C_j|$ is the number of generated coarse-grained candidates.

Fine-grained context generation. The goal of fine-grained context generation is to provide concise descriptions of mentioned entities in the target post. Short news posts frequently mention persons, organizations, or locations without sufficient explanation, which may hinder semantic interpretation. To address this issue, we design a reasoning-guided prompt that instructs the LLM to extract entities and provide stable and non-speculative descriptions for them.

Similar to the coarse-grained setting, the fine-grained prompt is also designed to mitigate hallucinations. Specifically, it requires the LLM to generate only objective information about entities explicitly mentioned in the post. Furthermore, it requires the LLM to rely solely on highly reliable and well-established knowledge when generating entity descriptions. Additionally, it requires the LLM to use conservative language when describing ambiguous entities. The prompt template is shown below.

Prompt for Fine-grained Context Generation

Task Description

You are a senior editor in the field of journalism. Given a news post from social media, your task is to: (1) identify entities mentioned in the news post (e.g., persons, organizations, or locations), and (2) provide concise descriptions for each entity that help readers understand its role or relevance in the news. The descriptions should be based on widely established knowledge and the given news. Follow the two-step process internally, but output only the final result.

Important requirements:

- Do not assume that the claim in the news post is true, and do not perform fact-checking or veracity judgment. Produce only objective information about entities. Do not include subjective opinions or personal speculation.
- Only describe entities that are explicitly mentioned in the post.
- For each entity, provide only highly reliable and widely known information.
- Do not add specific details that are not clearly supported by widely established knowledge. If an entity is ambiguous, use conservative and generic wording rather than guessing a specific identity, attribute, or relationship.

Input Data

Given news post: $[n_j]$

Output Format

Return the result in the following JSON format: $\{ "F_j": [f_{j,1}, f_{j,2}, \dots] \}$

Given a news post, we query the LLM with p_{FCG} to obtain fine-grained contextual candidates:

$$F_j = \text{LLM}(p_{FCG}(\tau_{FCG}, n_j, \varepsilon_{FCG})), \quad (2)$$

where $F_j = [f_{j,1}, f_{j,2}, \dots, f_{j,|F_j|}]$ denotes the fine-grained contextual candidates generated for the j th news post, p_{FCG} denotes the fine-grained prompt template, and $|F_j|$ is the number of generated fine-grained candidates.

By using reasoning-guided prompts with explicit anti-hallucination constraints, LMCE encourages the LLM to produce structured contextual candidates while reducing speculative and assumption-dependent outputs. These generated contexts are not treated as verified facts; instead, they serve as auxiliary contextual candidates for downstream selective fusion.

4.2.2. Encoder

After obtaining the multi-granularity contextual candidates, we employ a shared pre-trained BERT encoder (Devlin et al., 2019) to encode the original news text and each contextual candidate. The encoded news text is denoted by

$$\mathbf{n}_j = \text{BERT}(n_j), \quad (3)$$

where $\mathbf{n}_j \in \mathbb{R}^{d_t}$ is the semantic representation of the j th news post.

For the coarse-grained contextual candidates, we encode each candidate independently and stack their representations into a matrix:

$$\mathbf{c}_{j,m} = \text{BERT}(c_{j,m}), \quad \mathbf{C}_j = [\mathbf{c}_{j,1}; \mathbf{c}_{j,2}; \dots; \mathbf{c}_{j,|C_j|}] \in \mathbb{R}^{|C_j| \times d_t}. \quad (4)$$

Similarly, for the fine-grained contextual candidates, we have

$$\mathbf{f}_{j,m} = \text{BERT}(f_{j,m}), \quad \mathbf{F}_j = [\mathbf{f}_{j,1}; \mathbf{f}_{j,2}; \dots; \mathbf{f}_{j,|F_j|}] \in \mathbb{R}^{|F_j| \times d_t}. \quad (5)$$

4.2.3. Context enhancement

The contextual candidates generated by the LLM are not uniformly useful for fake news detection. Some candidates may be highly informative, while others may be weakly relevant or even carry hallucination risks. To address this issue, we design a cross-attention-based context enhancement module to selectively aggregate useful contextual candidates and integrate them with the original news semantics, thereby producing a context-enhanced semantic representation for the target news. Since coarse-grained and fine-grained context enhancement share the same cross-attention mechanism, we describe the coarse-grained branch in detail and then briefly introduce the fine-grained branch.

Coarse-grained context enhancement. In coarse-grained context enhancement, the news representation $\mathbf{n}_j$ serves as the query, while the coarse-grained context representations $\mathbf{C}_j$ are used as keys and values. The corresponding projections are defined as

$$\mathbf{Q}[n_j] = \mathbf{n}_j \mathbf{W}_{\text{CCE}}^Q, \quad \mathbf{K}[c_{j,m}] = \mathbf{c}_{j,m} \mathbf{W}_{\text{CCE}}^K, \quad \mathbf{V}[c_{j,m}] = \mathbf{c}_{j,m} \mathbf{W}_{\text{CCE}}^V, \quad (6)$$

where $\mathbf{W}_{\text{CCE}}^Q, \mathbf{W}_{\text{CCE}}^K, \mathbf{W}_{\text{CCE}}^V \in \mathbb{R}^{d_t \times d}$ are learnable projection matrices for coarse-grained context enhancement, and $\mathbf{c}_{j,m}$ denotes the representation of the m th coarse-grained contextual candidate associated with the j th news item.

The attention weight assigned to each coarse-grained contextual candidate is computed as

$$\text{att}_{\text{CCE}}(c_{j,m}, n_j) = \text{softmax}_{\forall c_{j,m} \in C_j} \left(\frac{\mathbf{Q}[n_j] \mathbf{K}[c_{j,m}]^T}{\sqrt{d}} \right). \quad (7)$$

Using these attention weights, we aggregate the coarse-grained contextual candidates and integrate them with the original news semantics to obtain the coarse-grained context-enhanced representation:

$$\mathbf{e}_{n_j}^{C \rightarrow n} = \sum_{\forall c_{j,m} \in C_j} \text{att}_{\text{CCE}}(c_{j,m}, n_j) \cdot \mathbf{V}[c_{j,m}] + \mathbf{n}_j \mathbf{W}_{\text{CCE}}, \quad (8)$$

where $\mathbf{W}_{\text{CCE}} \in \mathbb{R}^{d_i \times d}$ is a learnable projection matrix used to map the original news semantics into the same d -dimensional space as the aggregated contextual information. In this way, the model can emphasize useful contextual candidates while reducing the influence of noisy or hallucination-prone ones.

Fine-grained context enhancement. The fine-grained context enhancement follows the same cross-attention formulation, where the fine-grained entity-level context representations $\mathbf{F}_j$ are used as keys and values. Specifically, the resulting fine-grained context-enhanced news representation $\mathbf{e}_{n_j}^{F \rightarrow n}$ is obtained in the same manner as the coarse-grained branch. This branch allows the model to selectively incorporate reliable entity-level context and suppress entity descriptions that are less relevant or potentially misleading.

Finally, the coarse-grained and fine-grained context-enhanced representations are fused to produce the final context-enhanced semantic representation of the news:

$$\mathbf{e}_{n_j} = \text{FC}_{\text{fuse}} \left(\mathbf{e}_{n_j}^{C \rightarrow n} \parallel \mathbf{e}_{n_j}^{F \rightarrow n} \right), \quad (9)$$

where $\mathbf{e}_{n_j} \in \mathbb{R}^d$ denotes the final context-enhanced semantic representation of news n_j , FC_{fuse} is a single-layer fully connected network, and $\parallel$ denotes vector concatenation.

4.2.4. Knowledge distillation (Optional)

Although LMCE can provide semantically enriched news representations through LLM-generated contextual candidates, it requires querying an LLM for each news item during inference, which may be impractical in latency- or cost-sensitive deployment scenarios. To improve deployment efficiency, we further introduce an optional student model, termed LMCE-D, which distills the context enhancement capability of LMCE into a lightweight LLM-free module.

It should be noted that LMCE-D is not designed to explicitly recover complete world knowledge from sparse news text at inference time. Instead, its objective is to approximate the context-enhanced semantic representation space induced by the LLM-enhanced teacher, thereby retaining part of the context enhancement effect in a more efficient and deployable form.

Specifically, given a news item n_j , we first encode its text to obtain the base news representation $\mathbf{n}_j$, and then feed it into a lightweight context-aware feature simulator implemented by a transformer block. The simulator outputs a distilled representation $\mathbf{e}_{n_j}^D$, which is trained to approximate the teacher representation $\mathbf{e}_{n_j}$ produced by LMCE. The distillation objective is defined as

$$\mathcal{L}_{\text{MSE}} = \left\| \mathbf{e}_{n_j} - \mathbf{e}_{n_j}^D \right\|_2^2. \quad (10)$$

Through this objective, LMCE-D learns a mapping from sparse news text to LLM-enhanced semantic representations, rather than directly reproducing the teacher's external knowledge generation process. In this way, LMCE-D provides a compact approximation of the context-enhanced semantic representation without requiring LLM queries at inference time, thereby offering a favorable trade-off between semantic enhancement quality and deployment cost. Therefore, LMCE-D should be viewed as a practical approximation of teacher-induced semantic enhancement rather than a full replacement of the teacher's external knowledge generation capability.

4.3. Sample-specific cumulative propagation hypergraph construction

To explicitly model collective propagation behaviors while matching realistic deployment scenarios, we construct a sample-specific cumulative propagation hypergraph for each target news item. For a target news item n_j , let T_j denote its prediction time. We define the corresponding cumulative propagation hypergraph as

$$\mathcal{G}^{\leq T_j} = (\mathcal{U}^{\leq T_j}, \mathcal{N}^{\leq T_j}, \mathbf{A}^{\leq T_j}), \quad (11)$$

where $\mathcal{N}^{\leq T_j}$ denotes the set of news items published no later than T_j , $\mathcal{U}^{\leq T_j}$ denotes the set of users who have participated in propagating these news items by T_j , and $\mathbf{A}^{\leq T_j} \in \mathbb{R}^{|\mathcal{U}^{\leq T_j}| \times |\mathcal{N}^{\leq T_j}|}$ denotes the corresponding incidence matrix. Formally,

$$\mathcal{N}^{\leq T_j} = \left\{ n_z \mid t_z^{\text{pub}} \leq T_j \right\}, \quad (12)$$

$$\mathcal{U}^{\leq T_j} = \left\{ u_x \mid \exists n_z \in \mathcal{N}^{\leq T_j} \text{ such that } u_x \text{ participated in propagating } n_z \text{ at time } t_{x,z} \leq T_j \right\}, \quad (13)$$

and the incidence matrix is defined as

$$A_{x,z}^{\leq T_j} = \begin{cases} 1, & \text{if user } u_x \in \mathcal{U}^{\leq T_j} \text{ has participated in propagating } n_z \text{ by } T_j, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

This construction is sample-specific because the hypergraph is instantiated with respect to the target news item n_j , and it is cumulative because it aggregates all historically observable propagation information up to T_j . Therefore, the prediction of n_j is based only on the propagation evidence available at its prediction time, without accessing any future news item or future propagation record.

In this formulation, each user is treated as a node, and each news item is treated as a hyperedge connecting the users who participate in propagating it. This design preserves higher-order co-propagation relationships among multiple users involved in the same dissemination process. Unlike conventional graphs that decompose propagation into pairwise interactions, the constructed hypergraph explicitly models the collective news propagation process. Moreover, each user node is connected to the hyperedges corresponding to the news items that the user has historically propagated, enabling the model to capture user-level historical credibility cues from past propagation behaviors. Although the cumulative propagation hypergraph contains historical news items, this does not imply semantic homogeneity across different news items. Instead, different news items are linked only through overlapping users, indicating that the same user has participated in propagating them.

Hyperedge initialization. Each hyperedge is initialized by the semantic representation of its corresponding news item. In the LLM-enhanced setting, we initialize hyperedges by

$$\mathbf{H}_n^{(0)} = \mathbf{E}_n, \quad (15)$$

where $\mathbf{H}_n^{(0)} \in \mathbb{R}^{|\mathcal{N}^{\leq T_j}| \times d}$ denotes the initial hyperedge embedding matrix and $\mathbf{E}_n$ denotes the corresponding context-enhanced news representation matrix. In the distilled setting, we alternatively use the representation produced by LMCE-D:

$$\mathbf{H}_n^{(0)} = \mathbf{E}_n^D. \quad (16)$$

Node initialization. Existing propagation-based methods often initialize user nodes using profile attributes. However, such attributes are easily manipulated and may not reliably reflect actual user credibility. To mitigate this issue, we discard user profile information and instead infer user credibility from propagation behaviors. Specifically, user nodes are initialized via a learnable lookup table:

$$\mathbf{H}_u^{(0)} = \mathbf{W}_u [\mathcal{U}^{\leq T_j}], \quad (19)$$

where $\mathbf{H}_u^{(0)} \in \mathbb{R}^{|\mathcal{U}^{\leq T_j}| \times d}$ denotes the initial node embedding matrix, $\mathbf{W}_u \in \mathbb{R}^{|\mathcal{U}| \times d}$ is a trainable embedding matrix, $\mathcal{U}$ denotes the global user vocabulary, and d denotes the representation dimension.

4.4. Hypergraph transformer network

Collective propagation behavior patterns provide critical clues for distinguishing fake news from real news. Based on the constructed sample-specific cumulative propagation hypergraph, we propose a Hypergraph Transformer Network (HTN) to learn discriminative representations of collective propagation behavior patterns. The key idea of HTN is to design a hypergraph attention mechanism that captures higher-order correlations between nodes and hyperedges, as well as among nodes.

Specifically, based on the cumulative propagation hypergraph $\mathcal{G}^{\leq T_j} = (\mathcal{U}^{\leq T_j}, \mathcal{N}^{\leq T_j}, \mathbf{A}^{\leq T_j})$ of the target news item n_j , the l th HTN layer computes hypergraph attention and iteratively updates node and hyperedge representations. As illustrated in Fig. 2, the proposed hypergraph attention includes three levels: hyperedge-to-node attention, behavior-aware node-to-node attention, and node-to-hyperedge attention. Hyperedge-to-node attention aggregates connected hyperedge information to generate user representations that reflect global credibility. Behavior-aware node-to-node attention further produces news-specific user representations by incorporating propagation behavior signals and modeling behavioral influence among users participating in the same propagation process. Finally, node-to-hyperedge attention aggregates user representations within the target hyperedge to obtain the collective propagation behavior pattern representation of the target news item. For readability, we omit the superscript $\leq T_j$ in the following equations whenever no ambiguity arises.

4.4.1. Hyperedge-to-node attention

The hyperedge-to-node attention layer aims to learn a global credibility representation for each user by aggregating information from all news items connected to that user in the cumulative hypergraph. Intuitively, the global credibility of a user is jointly reflected by the set of news items that the user has historically propagated. For each user node, we employ an attention mechanism to selectively aggregate representations from its connected hyperedges. Specifically, we first compute the query, key, and value vectors as

$$\begin{aligned} Q[u_i] &= \mathbf{h}_{u_i}^{(l-1)} \mathbf{W}_{E \rightarrow V}^Q, \\ K[n_k] &= \mathbf{h}_{n_k}^{(l-1)} \mathbf{W}_{E \rightarrow V}^K, \\ V[n_k] &= \mathbf{h}_{n_k}^{(l-1)} \mathbf{W}_{E \rightarrow V}^V, \end{aligned} \quad (18)$$

where $\mathbf{W}_{E \rightarrow V}^Q, \mathbf{W}_{E \rightarrow V}^K, \mathbf{W}_{E \rightarrow V}^V \in \mathbb{R}^{d \times d}$ are trainable weight matrices. The attention coefficient of each connected hyperedge for user u_i is then computed as

$$\text{att}_{E \rightarrow V}(n_k, u_i) = \text{softmax}_{\forall n_k \in \mathcal{N}_i} \left(\frac{Q[u_i] K[n_k]^T}{\sqrt{d}} \right), \quad (19)$$

where $\mathcal{N}_i$ denotes the set of news hyperedges connected to user u_i in the cumulative hypergraph.

Finally, we aggregate hyperedge information according to the learned attention coefficients and combine it with the previous-layer user representation:

$$\mathbf{h}_{u_i}^{(l)} = \sigma \left(\sum_{\forall n_k \in \mathcal{N}_i} \text{att}_{E \rightarrow V}(n_k, u_i) V[n_k] \right) + \mathbf{h}_{u_i}^{(l-1)}. \quad (20)$$

This layer enables each user to acquire a global credibility representation by selectively aggregating information from connected news hyperedges.

4.4.2. Behavior-aware node-to-node attention

While hyperedge-to-node attention captures users' global credibility from historical propagation, collective propagation behavior patterns further depend on how users behave under a specific news context. For the same user, the role played in propagating different news items may vary substantially in terms of participation timing, structural position, and semantic reaction. Such news-specific behavioral differences are crucial for modeling collective propagation behavior patterns. To this end, we design a behavior-aware node-to-node attention module to generate news-specific user representations.

Propagation behavior encoding. For each user u_i participating in the propagation of news n_j , we explicitly model the user's observable propagation behavior under the target news context. Specifically, we encode three complementary behavior signals, namely, the temporal signal, the structural signal, and the comment-semantic signal. These signals jointly characterize when the user participates, how the user is structurally involved, and how the user semantically reacts during the propagation of n_j .

Temporal signal. The temporal signal characterizes the participation timing of user u_i in the propagation process of news n_j . Since collective propagation behavior patterns are closely related to the formation pace of the collective, capturing temporal participation is essential. We calculate the time difference between the timestamp $t_{i,j}$ when user u_i participates in propagating news n_j and the publication time t_j^{pub} of the source news, and then use an embedding function to obtain the temporal signal embedding:

$$\Delta t_{i,j} = t_{i,j} - t_j^{\text{pub}}, \quad \mathbf{s}_{i,j}^{\text{temp}} = \text{Emb}_{\text{temp}}(\Delta t_{i,j}), \quad (21)$$

where $\text{Emb}_{\text{temp}}(\cdot)$ denotes a learnable embedding function and $\mathbf{s}_{i,j}^{\text{temp}} \in \mathbb{R}^{d_s}$ denotes the temporal signal embedding.

Structural signal. The structural signal describes the structural role of user u_i in the propagation of n_j . We characterize it using two complementary features: (i) the propagation depth $d_{i,j}$ of user u_i , indicating its position along the diffusion path, and (ii) the number of child nodes $b_{i,j}$, reflecting its local propagation influence. We embed these two features and combine them to obtain the structural signal embedding:

$$\mathbf{s}_{i,j}^{\text{struct}} = \text{Emb}_{\text{struct}} \left(\text{Emb}_{\text{depth}}(d_{i,j}), \text{Emb}_{\text{child}}(b_{i,j}) \right), \quad (22)$$

where $\mathbf{s}_{i,j}^{\text{struct}} \in \mathbb{R}^{d_s}$ denotes the structural signal embedding, and $\text{Emb}_{\text{depth}}(\cdot)$, $\text{Emb}_{\text{child}}(\cdot)$, and $\text{Emb}_{\text{struct}}(\cdot)$ are learnable embedding functions.

Comment-semantic signal. The comment-semantic signal characterizes how user u_i semantically responds to news n_j during propagation. This signal is particularly important for preserving heterogeneous collective reactions, such as supportive, skeptical, debunking, or corrective responses. We first feed the user's comment into a pre-trained BERT encoder to extract its semantic features, and then use a trainable embedding function to obtain the comment-semantic signal embedding:

$$\mathbf{s}_{i,j}^{\text{com}} = \text{Emb}_{\text{com}}(\text{BERT}(c_{i,j})), \quad (23)$$

where $\mathbf{s}_{i,j}^{\text{com}} \in \mathbb{R}^{d_s}$ denotes the comment-semantic signal embedding.

Finally, we fuse the global user representation with the above three behavior signals to obtain the behavior-aware user representation:

$$\hat{\mathbf{h}}_{u_i}^{(l)} = \mathbf{W}_{\text{behav}} \left(\mathbf{h}_{u_i}^{(l)} \parallel \mathbf{s}_{i,j}^{\text{temp}} \parallel \mathbf{s}_{i,j}^{\text{struct}} \parallel \mathbf{s}_{i,j}^{\text{com}} \right), \quad (24)$$

where $\hat{\mathbf{h}}_{u_i}^{(l)} \in \mathbb{R}^d$ denotes the behavior-aware user representation and $\mathbf{W}_{\text{behav}} \in \mathbb{R}^{d \times (d+3d_s)}$ is a trainable parameter matrix. In this way, each user is represented not only by global historical credibility, but also by its news-specific propagation behavior under the target news context.

Node-to-node attention. After obtaining behavior-aware user representations, we further model the behavioral interaction among users participating in the same target news. The goal of node-to-node attention is to refine each user's representation by modeling how other users' propagation behaviors influence the target user's role within the same collective.

Specifically, for a target user u_i , we compute the query, key, and value vectors as

$$\begin{aligned} Q[u_i] &= \hat{\mathbf{h}}_{u_i}^{(l)} \mathbf{W}_{V \rightarrow V}^Q, \\ K[u_p] &= \hat{\mathbf{h}}_{u_p}^{(l)} \mathbf{W}_{V \rightarrow V}^K, \\ V[u_p] &= \hat{\mathbf{h}}_{u_p}^{(l)} \mathbf{W}_{V \rightarrow V}^V, \end{aligned} \quad (25)$$

where $\mathbf{W}_{V \rightarrow V}^Q, \mathbf{W}_{V \rightarrow V}^K, \mathbf{W}_{V \rightarrow V}^V \in \mathbb{R}^{d \times d}$ are trainable weight matrices. The interaction coefficient from user u_p to target user u_i is computed as

$$\text{att}_{V \rightarrow V}(u_p, u_i) = \text{softmax}_{\forall u_p \in \mathcal{U}_j} \left(\frac{Q[u_i]K[u_p]^T}{\sqrt{d}} \right), \quad (26)$$

where $\mathcal{U}_j$ denotes the set of users participating in propagating the target news item n_j .

Then, the refined news-specific user representation is updated as

$$\mathbf{h}_{u_i, j}^{(l)} = \sigma \left(\sum_{\forall u_p \in \mathcal{U}_j} \text{att}_{V \rightarrow V}(u_p, u_i) V[u_p] \right) + \hat{\mathbf{h}}_{u_i, j}^{(l)}. \quad (27)$$

Through node-to-node attention, the model obtains news-specific user representations that preserve user-level behavioral heterogeneity and provide behavior-aware inputs for subsequent collective-level aggregation.

4.4.3. Node-to-hyperedge attention

After obtaining news-specific user representations, we aggregate the heterogeneous users and their individual behaviors into a hyperedge representation by node-to-hyperedge attention. Its objective is to learn a representation that characterizes the collective propagation behavior pattern of the target news item.

Collective propagation behavior patterns are inherently macro-level patterns. They are not determined by any single user, but by the combined temporal participation, structural involvement, and comment-semantic responses of all participating users. Benefiting from the flexible hyperedge structure, node-to-hyperedge attention can adaptively integrate these heterogeneous user-level behaviors into an event-level propagation pattern representation.

Specifically, we first compute the query, key, and value vectors:

$$\begin{aligned} Q[n_j] &= \mathbf{h}_{n_j}^{(l-1)} \mathbf{W}_{V \rightarrow E}^Q, \\ K[u_i] &= \mathbf{h}_{u_i, j}^{(l)} \mathbf{W}_{V \rightarrow E}^K, \\ V[u_i] &= \mathbf{h}_{u_i, j}^{(l)} \mathbf{W}_{V \rightarrow E}^V, \end{aligned} \quad (28)$$

where $\mathbf{W}_{V \rightarrow E}^Q, \mathbf{W}_{V \rightarrow E}^K, \mathbf{W}_{V \rightarrow E}^V \in \mathbb{R}^{d \times d}$ are trainable weight matrices. The node-to-hyperedge attention coefficient is calculated as

$$\text{att}_{V \rightarrow E}(u_i, n_j) = \text{softmax}_{\forall u_i \in \mathcal{U}_j} \left(\frac{Q[n_j]K[u_i]^T}{\sqrt{d}} \right). \quad (29)$$

Finally, the updated hyperedge representation is obtained by

$$\mathbf{h}_{n_j}^{(l)} = \sigma \left(\sum_{\forall u_i \in \mathcal{U}_j} \text{att}_{V \rightarrow E}(u_i, n_j) V[u_i] \right) + \mathbf{h}_{n_j}^{(l-1)}. \quad (30)$$

Through node-to-hyperedge attention, the model adaptively aggregates different users together with their individual propagation behaviors, yielding a hyperedge representation that characterizes the collective propagation behavior pattern of the target news item. In particular, this representation is determined by the combined temporal participation, structural involvement, and comment-semantic responses of all participating users.

For notational compactness, we denote the above three-level hypergraph attention block on the sample-specific cumulative propagation hypergraph by $\text{HyA}(\cdot)$, and extend it to a multi-head version as follows.

$$\begin{aligned} \text{MH-HyA}(\mathbf{H}_n^{(l-1)}, \mathbf{A}) &= \sigma \left(\mathbf{W}_{MH} \left\|_{m=1}^M \text{head}_m \right\| \right), \\ \text{head}_m &= \text{HyA}_m(\mathbf{H}_n^{(l-1)}, \mathbf{A}), \end{aligned} \quad (31)$$

where $\sigma(\cdot)$ denotes the activation function, M is the number of attention heads, $\mathbf{W}_{MH}$ is the learnable projection matrix for multi-head aggregation, and $\|$ denotes vector concatenation.

Following the standard transformer design, we further employ a feed-forward network (FFN) together with residual connections and layer normalization (LN) to stabilize training and preserve representation consistency across layers. The update process is formulated as

$$\begin{aligned} \mathbf{H}_n^{(l)} &= \text{LN}(\text{MH-HyA}(\mathbf{H}_n^{(l-1)}, \mathbf{A}) + \mathbf{H}_n^{(l-1)}), \\ \mathbf{H}_n^{(l)} &= \text{LN}(\text{FFN}(\mathbf{H}_n^{(l)}) + \mathbf{H}_n^{(l)}). \end{aligned} \quad (32)$$

By stacking L hypergraph transformer layers, the proposed model progressively captures complex higher-order correlations in collective propagation behavior and yields expressive collective propagation behavior pattern representations for fake news detection.

After producing collective propagation behavior pattern representations through the hypergraph transformer network module, we use a single-layer classifier to obtain the classification results of the news.

$$\hat{y}_i = \text{Softmax} \left(\mathbf{h}_{n_i}^{(L)} \mathbf{W}_{\text{cls}} + \mathbf{b}_{\text{cls}} \right) \quad (33)$$

Table 1
Dataset statistics.

Datasets	Users	News	Labels	Language	Social platform
Twitter	29,858	1152	R:577/F:575	English	Twitter
PHEME	25,575	1705	R:1067/F:638	English	Twitter
DRWeibo	98,491	6037	R:3185/F:2852	Chinese	Weibo

where $\mathbf{W}_{\text{cls}}$ and $\mathbf{b}_{\text{cls}}$ are the learnable parameters. For the fake news detection task, we train the model using the standard cross-entropy loss function as below:

$$\mathcal{L}_{CE} = -\frac{1}{|\mathcal{N}|} \sum_{i=1}^{|\mathcal{N}|} y_i \cdot \log \hat{y}_i \quad (34)$$

where $|\mathcal{N}|$ is the number of news, y_i and $\hat{y}_i$ respectively denote the ground-truth label and the predicted label of the i th news from the classifier.

5. Experiment

In this section, we evaluate the performance of our proposed LeHT model on three datasets and answer the following research questions:

- **RQ1.** How does the proposed LeHT perform on the fake news detection task compared to existing SOTA methods?
- **RQ2.** Can LeHT identify fake news at the early stage of propagation?
- **RQ3.** How does the proposed innovation strategy contribute to the performance of LeHT?
- **RQ4.** How sensitive is the LeHT model to LLM selection and hyperparameter settings?
- **RQ5.** How time-efficient is the LeHT model?
- **RQ6.** Are the collective propagation behavior pattern features we focus on truly effective for detecting fake news?
- **RQ7.** Can large language models generate the contextual information necessary for comprehending news semantics?

5.1. Experimental setup

5.1.1. Dataset

We evaluate the proposed LeHT on three widely used public datasets: Twitter, PHEME, and DRWeibo. Table 1 summarizes their statistics. Detailed descriptions are provided as follows:

- **Twitter** dataset introduced by Ma et al. (2017) is a widely used English dataset for fake news/rumor detection. It is collected from Twitter¹ and provides both the textual content of each news item and its corresponding propagation records. The original dataset includes four categories: *non-rumor*, *unverified rumor*, *true rumor*, and *false rumor*. Since our task is binary fake news detection, following prior work (Song et al., 2022), we treat *true rumor* as real news and *false rumor* as fake news, and remove the *non-rumor* and *unverified rumor* categories. After processing, the statistics of the dataset are shown in Table 1.
- **PHEME** dataset is released by Kochkina et al. (2018) and is a widely adopted English fake news dataset. The dataset is collected based on nine breaking events from the Twitter² platform. The PHEME dataset contains news content together with associated propagation data, where each item is annotated as *real*, *fake*, or *unverified* by professional annotators. Since the *unverified* category lacks definitive veracity labels, we retain only the *real* and *fake* items for training and evaluation. After processing, summary statistics are provided in Table 1.
- **DRWeibo** dataset is proposed by Cui and Jia (2024) and is widely utilized in the Chinese fake news detection task. The DRWeibo dataset comprises Chinese news items along with their propagation data. The dataset is collected from Weibo,³ a major social media platform in China. Each item is labeled as *real* or *fake* by professional annotators. The statistics of DRWeibo are summarized in Table 1.

5.1.2. Comparative methods

To comprehensively evaluate the effectiveness of LeHT, we compare it with state-of-the-art fake news detection methods, covering content-based approaches and propagation-based approaches. Below is a brief introduction to the comparative methods.

¹ <https://x.com>.

² <https://twitter.com/>.

³ <https://weibo.com>.

Content-based methods.

- End-to-end Methods:
ML-GRU (Ma et al., 2016) employs a multi-layer GRU to encode news texts for classification.
CAMI (Yu et al., 2017) extracts semantic features using a convolutional neural network from news text content.
VROC (Cheng et al., 2020) incorporates an autoencoder with a multi-task learning mechanism to enhance semantic understanding.
- PLM-based Methods:
BERT (Devlin et al., 2019) is a well-known pre-trained language model. We fine-tune it for fake news detection.
DeBERTa (He et al., 2021) is a popular pre-trained language model. We fine-tune it on the downstream task dataset for fake news detection.
- LLM-based Methods:
LLaMA-3 (Touvron et al., 2023) is an open-source large language model developed by Meta. We employ the Llama-3.1-8B-Instruct version.
GPT-3.5 (OpenAI, 2022) is a closed-source large language model proposed by OpenAI. We use the prompt learning technique to stimulate the potential of GPT-3.5 for fake news detection.
GPT-4.0 (Achiam et al., 2023) is an upgraded version of GPT-3.5, which is a popular large language model.
ARG (Hu et al., 2024) utilizes the LLM to generate multi-perspective rationales related to news veracity and uses a small model to selectively derive news analysis insights from LLM rationales.
LEKD (Chen et al., 2025) utilizes LLMs to generate rationales for judging news authenticity and proposes a graph-based semantic feature alignment module to obtain news representations for classification.
NRFE (Zhang et al., 2025) employs LLMs to produce positive and negative rationales for proving the veracity of news and designs a semantic consistency learning module to align the semantics between news and rationales.

Propagation-based methods.

- Simple Graph-based Methods:
TRM-CPM (Li et al., 2020) uses GraphSAGE to encode propagation structures.
Bi-GCN (Bian et al., 2020) leverages a top-down and a bottom-up GCN to learn the patterns of rumor propagation and rumor dispersion, respectively.
GMVCN (Wu et al., 2024) uses a bi-directional GCN to model multi-view propagation information of news and uses CNN to fuse multi-view information.
SPGAT (L. Li et al., 2025) uses social psychology information to enhance node attributes and utilizes a bidirectional graph attention network to learn news propagation representations.
EASE-GARD (G. Li et al., 2025) is an emotion-aware structural enhancement graph autoencoder framework to learn emotion-enriched propagation representations for fake news detection.
RAGCL (Cui & Jia, 2024) uses node centrality to guide adaptive data augmentation and proposes adaptive graph contrastive learning for fake news detection.
HPCL (Zheng et al., 2024) proposes a hierarchical prototype contrastive learning method to train the GNN for fake news detection.
AD-GSCL (Cui & Jia, 2025) incorporates supervised graph contrastive learning and unsupervised graph contrastive learning to optimize graph neural networks, leading to highly discriminative news representations.
- Dynamic Graph-based Methods
DGNF (Song et al., 2022) models propagation as discrete graph snapshots. It uses GAT to capture propagation structure information in each snapshot and uses a self-attention mechanism to capture the temporal dependencies between propagation snapshots.
MST (Huang et al., 2025) constructs dual dynamic graph convolutional networks to simultaneously learn spatio-temporal features of both information propagation and user interaction, effectively fusing content, propagation, and user features.
SWAM (Guo et al., 2025b) is an adaptive sliding window and memory-augmented attention model for fake news detection. It proposes an adaptive sliding window mechanism that partitions the news propagation process into a sequence of disjoint windows and introduces a memory-enhanced attention method to capture the temporal information of news propagation.
FGDGNN (Guo et al., 2025a) proposes a fine-grained dynamic graph neural network to capture fine-grained temporal evolution information in the process of news dissemination, including edge-aware temporal information and node-level dynamic update information.
- Hypergraph-based Methods
SA-HyperGAT (Dong et al., 2022) applies a sentiment-aware hypergraph attention network based on a dual attention mechanism to learn higher-order relations among comments with the same sentimental polarity.
CBT-HGCN (Li et al., 2023) uses a hypergraph to model higher-order correlations among retweets in the conversation branch and uses hypergraph convolutional networks to learn conversation representations for fake news detection.
Hy-DeFake (Su et al., 2025) utilizes a hypergraph neural network to model higher-order correlations between users and their historical retweets, thereby capturing the global credibility representation of the user. It then concatenates the user credibility representations and the news content representation for fake news detection.

HG-SL (Sun et al., 2023) designs a hypergraph neural network module to capture the global preferences of users from their spreading behaviors and a self-attention-based local context learning module to capture temporal and structural information during the news propagation process.

Model variants. To make a comprehensive comparison, we evaluate two variants of our framework in the experiments, namely LeHT and LeHT-D.

- **LeHT** is the full model, which integrates an LLM-driven multi-granularity context enhancement (LMCE) module with a hypergraph transformer network (HTN) module to capture collective-level propagation patterns. In this setting, the LLM is queried to generate contextual information that enriches the semantic representation of each news post.
- **LeHT-D** is a distilled variant designed for cost-sensitive and efficient deployment. This variant employs the knowledge-distilled LMCE-D model to generate context-enhanced news semantic representations.

5.1.3. Implementation details

All experiments were conducted on a server equipped with two NVIDIA RTX 4090 GPUs. All baselines and our proposed model were implemented in PyTorch.⁴ To better match realistic deployment scenarios, we adopted a strict temporal split strategy. Specifically, all news items were sorted according to their publication timestamps. The earliest 80% of news items were used for training, while the latest 20% were used for testing. Under this temporal split strategy, the model is trained on past news and evaluated on future news, which is more consistent with realistic deployment scenarios.

For our proposed LeHT model, we constructed a sample-specific cumulative propagation hypergraph for each news item. Since the training news items are published earlier than the test news items, historically observable propagation records prior to the prediction time of each target item can be used. However, during training, the model can never access any test-period news item or its associated propagation records.

The implementation details of our proposed LeHT model are as follows. In the LLM-driven multi-granularity context enhancement (LMCE) module, we used GPT-4.0 to generate multi-granularity contextual information. For text encoding, we adopted `bert-base-chinese`⁵ for the Chinese dataset DRWeibo, and `bert-base-uncased`⁶ for the English datasets Twitter and PHEME. The output embedding dimension of BERT was set to $d_i = 768$. In the Hypergraph Transformer Network (HTN) module, we set the dimension of both node and hyperedge embeddings to $d = 256$, the dimension of signal embeddings to $d_s = 64$, the number of hypergraph attention heads to $h = 8$, and the number of hypergraph transformer layers to $L = 2$. During training, we set the learning rate to 0.01, the batch size to 64, and the number of epochs to 100, and used Adam (Kingma & Ba, 2015) as the optimizer.

For comparative methods, we used the hyperparameter settings reported in the original papers to ensure their best performance on each dataset. For pre-trained language models (BERT and DeBERTa), we fine-tuned them on the corresponding training split of each dataset to adapt them to the downstream task. For large language models (LLaMA-3, GPT-3.5, and GPT-4.0), following prior work (Raza et al., 2025), we used prompting to obtain veracity predictions for fake news detection. For all comparative methods, we re-implemented them or ran their official code under the same temporal split strategy whenever possible. The prompt template is provided below.

Prompt for detecting fake news using LLMs

```
# Task Description
You are a professional expert in fake news detection. Your task is to determine the veracity
of a given news post based on its content. You must provide a definitive judgment and avoid
ambiguous responses.
# Input Data
Given news post: [nj]
# Output Format
Return a single label:
- 1, if the news is more likely to be real. - 0, if the news is more likely to be fake.
Only output the label (1 or 0). Do not provide any explanation, justification, or additional
text.
```

For a fair and stable evaluation under the temporal split strategy, we fixed the data split and repeated each experiment five times with different random seeds. We report the average ACC and F1-score over these repeated runs. To examine whether the performance improvement of LeHT over the best comparative method is statistically significant, we further conducted paired *t*-tests based on the results of repeated runs. The improvement is regarded as statistically significant when the corresponding *p*-value is below 0.05.

⁴ <https://pytorch.org/>.

⁵ <https://huggingface.co/google-bert/bert-base-chinese>.

⁶ <https://huggingface.co/google-bert/bert-base-uncased>.

Table 2

Results of comparison between LeHT and other fake news detection methods on three datasets. We evaluate the performance of the model using ACC and F1. Bold numbers in the table indicate the best performance.

		Method	Twitter ACC	F1	PHEME ACC	F1	DRWeibo ACC	F1
Content-based methods	End-to-end methods	ML-GRU (Ma et al., 2016)	0.645	0.642	0.651	0.648	0.644	0.642
		CAMI (Yu et al., 2017)	0.652	0.647	0.663	0.652	0.646	0.642
		VROC (Cheng et al., 2020)	0.663	0.659	0.661	0.663	0.664	0.661
	PLM-based methods	BERT (Devlin et al., 2019)	0.704	0.701	0.717	0.716	0.724	0.719
		DeBERTa (He et al., 2021)	0.699	0.702	0.712	0.714	0.729	0.728
	LLM-based methods	LLaMA-3 (Touvron et al., 2023)	0.603	0.598	0.612	0.609	0.592	0.591
		GPT-3.5 (OpenAI, 2022)	0.622	0.618	0.619	0.618	0.618	0.617
		GPT-4.0 (Achiam et al., 2023)	0.635	0.627	0.628	0.631	0.626	0.631
		ARG (Hu et al., 2024)	0.749	0.753	0.761	0.763	0.767	0.762
		LEKD (Chen et al., 2025)	0.753	0.756	0.768	0.762	0.754	0.761
		NRFE (Zhang et al., 2025)	0.764	0.761	0.759	0.754	0.759	0.760
Propagation-based methods	Simple graph-based methods	TRM-CPM (Li et al., 2020)	0.794	0.789	0.787	0.792	0.794	0.791
		Bi-GCN (Bian et al., 2020)	0.804	0.801	0.783	0.775	0.802	0.798
		GMVCN (Wu et al., 2024)	0.817	0.821	0.805	0.794	0.827	0.816
		SPGAT (L. Li et al., 2025)	0.813	0.816	0.811	0.807	0.826	0.824
		EASE-GARD (G. Li et al., 2025)	0.831	0.824	0.814	0.812	0.834	0.830
		RAGCL (Cui & Jia, 2024)	0.835	0.832	0.814	0.816	0.836	0.829
		HPCL (Zheng et al., 2024)	0.824	0.826	0.827	0.825	0.836	0.831
		AD-GSCL (Cui & Jia, 2025)	0.831	0.834	0.825	0.821	0.832	0.833
	Dynamic graph-based methods	DGNF (Song et al., 2022)	0.847	0.836	0.841	0.844	0.844	0.841
		MST (Huang et al., 2025)	0.852	0.851	0.847	0.841	0.859	0.851
		SWAM (Guo et al., 2025b)	0.867	0.859	0.869	0.862	0.866	0.871
		FGDGNN (Guo et al., 2025a)	0.882	0.885	0.873	0.879	0.868	0.869
	Hypergraph-based methods	SA-HyperGAT (Dong et al., 2022)	0.854	0.849	0.882	0.847	0.862	0.858
		CBT-HGCN (Li et al., 2023)	0.855	0.852	0.841	0.832	0.867	0.864
		Hy-DeFake (Su et al., 2025)	0.847	0.842	0.843	0.847	0.856	0.854
		HG-SL (Sun et al., 2023)	0.875	0.871	0.884	0.877	0.874	0.871
	Our methods	LeHT-D	0.918*	0.914*	0.921*	0.919*	0.911*	0.907*
		LeHT	0.923*	0.921*	0.926*	0.924*	0.916*	0.913*

* Demonstrates that the p -value is below 0.05 using the paired t -test over repeated runs compared with the best comparative method.

5.2. Performance comparison with state-of-the-art methods (RQ1)

We compare the performance of LeHT with other state-of-the-art fake news detection methods on each dataset. The results of the comparison are displayed in Table 2. We can draw the following conclusions.

- Our proposed LeHT outperforms all comparative methods on the three datasets in terms of both ACC and F1-score. Specifically, LeHT achieves accuracies of 0.923, 0.926, and 0.916 on Twitter, PHEME, and DRWeibo, respectively. We further conduct paired t -tests between LeHT and the best-performing baseline on each dataset and obtain p -values below 0.05, indicating that the improvements are statistically significant. These results suggest that the collective-level propagation analysis perspective provides effective signals for fake news detection. We attribute the superiority of LeHT to two main factors: (i) the LLM-driven multi-granularity context enhancement (LMCE) module leverages the knowledge and semantic capabilities of LLMs to generate multi-granularity contextual information, which mitigates semantic sparsity in short posts and helps capture semantic cues that may trigger collective propagation; and (ii) the proposed hypergraph transformer network (HTN) module can effectively capture higher-order correlations in collective propagation behaviors and generate highly discriminative representations of collective propagation behavior patterns.
- Overall, content-based methods underperform propagation-based approaches on three datasets, suggesting that text alone is often insufficient to distinguish fake news that is carefully crafted to resemble real news. Within content-based methods, pretrained language model (PLM) approaches (BERT and DeBERTa) generally outperform end-to-end models (e.g., ML-GRU, CAMI, and VROC), indicating that large-scale pretraining improves semantic representation and understanding. However, directly using LLMs (LLaMA-3, GPT-3.5, and GPT-4.0) in zero-shot fake news detection yields unsatisfactory performance and can even be inferior to end-to-end methods in some settings. This is because the lack of task-specific adaptation in the zero-shot setting limits the model's ability to extract fake cues. In contrast, methods that treat LLMs as advisors and combine them with smaller task-specific models (e.g., ARG, LEKD, and NRFE) typically outperform other content-based methods. This observation implies that integrating the language processing strengths of LLMs with the task-specific discriminative ability of small models can be a more effective strategy. Inspired by this, LeHT uses LLM-generated contextual information while training a downstream model to learn semantic cues.

- Propagation-based methods outperform content-based methods, indicating that incorporating propagation information is essential for detecting fake news. Simple graph-based approaches leverage graph neural networks to capture structural information in news propagation, achieving some effectiveness. Furthermore, dynamic graph-based methods can simultaneously capture both temporal and structural information within the news propagation process. Consequently, most dynamic graph-based approaches outperform simple graph-based ones. However, constrained by the pairwise connection nature of conventional graphs, these methods struggle to capture higher-order correlations in collective propagation behavior, thus limiting their ability to achieve optimal results.

We can see that LeHT achieves superior performance compared with existing hypergraph-based methods. We attribute this to two aspects: (1) Motivated by collective-level propagation analysis, LeHT designs a hypergraph transformer network that explicitly models higher-order correlations in news propagation data and learns more discriminative macro-level collective propagation behavior pattern representations. In contrast, existing hypergraph methods often focus on modeling higher-order relations within specific interaction units in the propagation process (e.g., sentiment-consistent comments, comments within the same conversation branch, or posts associated with the same user), which can be insufficient to capture the overall collective propagation process. (2) LeHT incorporates LLM-generated multi-granularity contexts to enrich news semantics, whereas most hypergraph-based methods do not explicitly address the semantic sparsity of short news posts.

5.3. Fake news early detection performance comparison with state-of-the-art methods (RQ2)

Fake news early detection is crucial for safeguarding information security and social stability. Its objective is to determine the authenticity of a news item at the early stage of propagation, thereby mitigating the harm caused by large-scale diffusion. However, early detection can be evaluated from different perspectives depending on what is constrained during observation. In realistic online scenarios, a detector often needs to make decisions within a fixed time budget. At the same time, under the same detection deadline, different news items may expose substantially different amounts of propagation evidence due to heterogeneous activity levels and observation densities. To provide a more comprehensive evaluation, we consider two complementary early detection scenarios in this work, namely, the deadline-limited scenario and the engagement-limited scenario.

Deadline-limited scenario. Following prior studies (Guo et al., 2025a, 2025b), we first evaluate early detection under a series of detection deadlines, i.e., 0, 20, 40, 60, 80, 100, and 120 min. For each deadline, we truncate the observed propagation records up to the corresponding time window and evaluate the detection performance. This scenario reflects a realistic time-constrained setting, where the detector must make a decision within a fixed temporal budget. The ACC results of LeHT and representative early-detection baselines, including Bi-GCN, FGDGNN, SWAM, HG-SL, and Hy-DeFake, are reported in Fig. 3.

As shown in Fig. 3, when the deadline is 0 min, all methods achieve relatively poor performance. This is because propagation data are unavailable at the beginning, and the models have to rely only on content information. As the deadline increases, the performance of all methods improves, benefiting from richer propagation information. Notably, LeHT outperforms all baselines across all deadlines. Moreover, LeHT already achieves over 90% accuracy at the 120-minute deadline, demonstrating strong early detection capability. This advantage is attributed to LeHT's effective modeling of collective-level propagation behavior patterns. During the early propagation stages, fake news may exhibit more coordinated propagation behavior (e.g., users with similar motivations retweeting in high synchrony within a short timeframe), thereby generating more pronounced collective propagation signals. By explicitly modeling higher-order relationships in collective propagation through propagation hypergraphs, LeHT can more fully leverage these early signals, leading to superior performance in early detection tasks.

Engagement-limited scenario. Under the same detection deadline, different news items may expose substantially different amounts of propagation records due to heterogeneous activity levels and observation densities. To reduce the influence of heterogeneous observation density under the same deadline, we further introduce an engagement-limited scenario. In this setting, for each news item, we retain only the earliest 20, 40, 60, 80, or 100 retweets. By fixing the amount of observed propagation evidence across news items, this scenario provides a more density-controlled comparison of early detection performance. Under this setting, we again compare LeHT with the early-detection baselines, and the results are reported in Fig. 4.

As shown in Fig. 4, LeHT also maintains superior performance across different engagement budgets. This result is important because it shows that the advantage of LeHT does not merely come from observing denser or more active propagation under a fixed deadline. Instead, even when the amount of observed propagation evidence is controlled, LeHT can still better exploit limited early propagation signals. This further confirms that the proposed framework effectively captures collective propagation behavior patterns from sparse early-stage observations.

The two scenarios are complementary. The deadline-limited scenario evaluates practical early detection ability under realistic temporal constraints, whereas the engagement-limited scenario controls the amount of observed propagation records. The results under both scenarios demonstrate that LeHT achieves strong early fake news detection performance. The deadline-limited scenario verifies its practical utility under realistic time constraints, while the engagement-limited scenario provides additional evidence that its improvement is not simply due to differences in observation density, but rather to its stronger ability to model collective propagation behavior patterns.

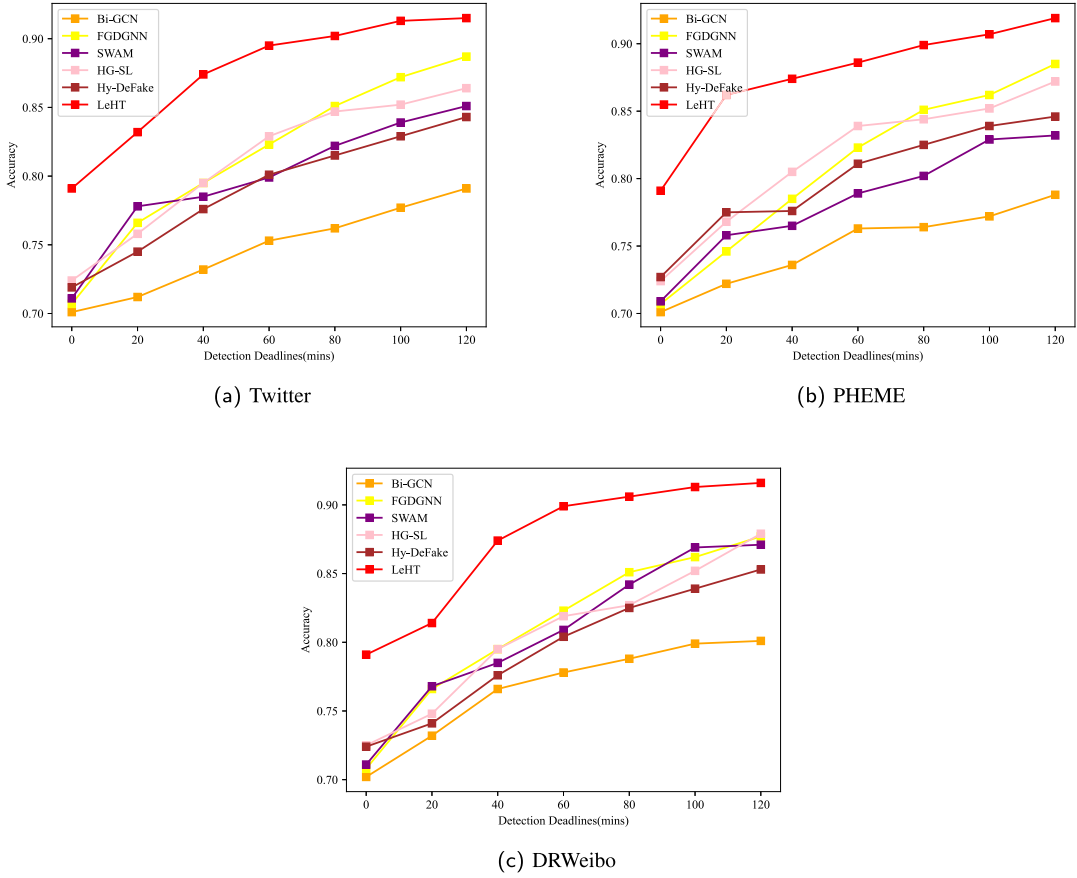


Fig. 3. Results of deadline-limited early fake news detection on three datasets.

5.4. Ablation study (RQ3)

In this subsection, we conduct a comprehensive ablation study to quantify the contribution of each proposed innovation in LeHT. We first examine the effectiveness of the major model components. Then, we further analyze the effectiveness of different context enhancement designs, including the contribution of coarse-grained and fine-grained context, as well as the advantage of cross-attention-based context enhancement over naive pooling.

5.4.1. Effectiveness of each component

To verify the effectiveness of each component, we design the following variants of LeHT:

- **LMCE**: We remove the hypergraph Transformer network (HTN) module and perform fake news detection using only the context-enhanced news representations generated by the LMCE module.
- **LMCE-D**: We remove the hypergraph Transformer network (HTN) module and perform fake news detection using only the context-enhanced news representations generated by the knowledge-distilled LMCE-D module.
- **BERT+HTN**: We remove the LLM-driven multi-granularity context enhancement module (LMCE). Instead, we encode the raw news content using the BERT model and feed the news content representation into HTN for fake news detection.
- **LMCE+GTN**: We replace HTN with a graph transformer network (GTN) that models the propagation structure using pairwise connected edges, while keeping LMCE for semantic enhancement. It is worth noting that the graph transformer network is currently a powerful backbone architecture for graph representation learning.
- **LMCE+HGCN**: We replace HTN with a hypergraph convolutional network (HGCN) to learn collective propagation behavior pattern representations on the constructed propagation hypergraph, while keeping LMCE for semantic enrichment.

We compare LeHT with its variants on three datasets, and the results are reported in Table 3. We can draw the following conclusions from the experiment results: 1. LeHT achieves the best performance on all three datasets, indicating that each component contributes positively to the final effectiveness. 2. Both LMCE and LMCE-D underperform the full LeHT. This indicates that capturing collective propagation behavior patterns is crucial for detecting fake news. Relying solely on content is unlikely to achieve optimal

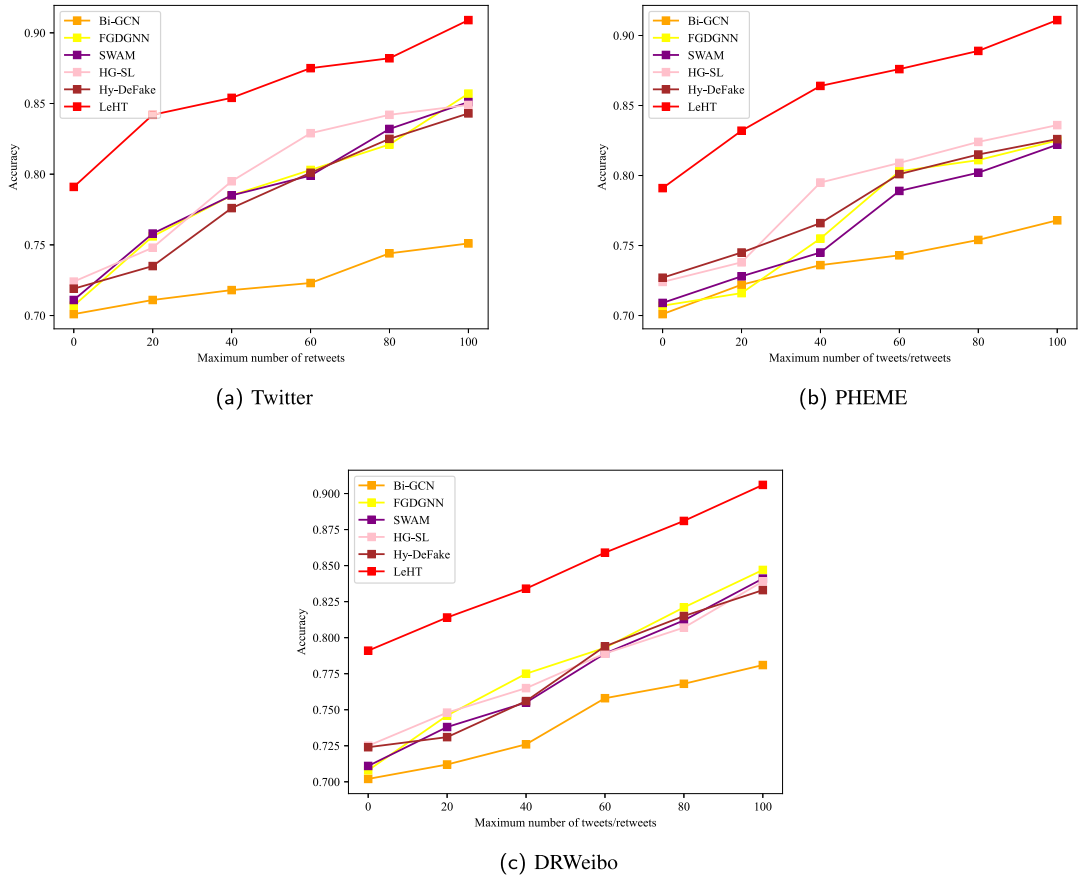


Fig. 4. Results of engagement-limited early fake news detection on three datasets.

Table 3

Results of comparison between LeHT and its variant models on the three datasets. Bold numbers in the table indicate the best performance.

Method	Twitter		PHEME		DRWeibo	
	ACC	F1	ACC	F1	ACC	F1
LeHT	0.923	0.921	0.926	0.924	0.916	0.913
Effectiveness of each component of the model						
LMCE	0.791	0.788	0.791	0.792	0.788	0.793
LMCE-D	0.774	0.773	0.776	0.787	0.785	0.786
BERT+HTN	0.907	0.908	0.909	0.906	0.896	0.898
LMCE+GTN	0.852	0.848	0.847	0.842	0.846	0.848
LMCE+HGCN	0.878	0.872	0.866	0.869	0.864	0.861
Effectiveness of context enhancement design						
LeHT w/o CCE	0.917	0.913	0.919	0.916	0.906	0.905
LeHT w/o FCE	0.916	0.911	0.916	0.912	0.907	0.904
LeHT w/ pooling CE	0.914	0.912	0.918	0.911	0.905	0.903

results. 3. Compared with content-based fake news detection approaches, LMCE achieves better results, implying that multi-granularity context enhancement can effectively mitigate semantic sparsity in short news posts and improve the comprehension of news semantics. Moreover, LMCE-D remains reasonably competitive with LMCE, demonstrating that the knowledge distillation strategy can preserve most of the enhancement capability in a lightweight model. 4. BERT+HTN performs worse than LeHT on all datasets, demonstrating the effectiveness of the LMCE module in capturing news semantic features. BERT only extracts features from news text. However, due to the limited length of news articles, their semantic information is sparse. In contrast, the LMCE module leverages the rich world knowledge of LLM to enrich the semantics of news posts. We found that even without using LLM-enhanced semantic representations, BERT+HTN still achieved strong performance, which indicates that collective-level propagation

modeling makes an important contribution. 5. LMCE+GTN underperforms LeHT, suggesting that even with a strong graph backbone network, modeling propagation with pairwise edges is still insufficient for fully capturing higher-order collective propagation behaviors. In contrast, the hypergraph formulation in LeHT preserves the collective semantics of multiple users participating in the same news propagation process, and is therefore better suited to collective-level propagation analysis. 6. LMCE+HGCN also performs worse than LeHT, which further clarifies that the gains of LeHT do not come merely from using hyperedges as a more aggregated representation. Importantly, LMCE+HGCN shares the same LMCE module and the same hypergraph construction as LeHT, so both models equally benefit from hyperedge-based higher-order aggregation. However, HGCN mainly passes messages between nodes and hyperedges with fixed weights, yielding a relatively generic form of hypergraph aggregation. It does not explicitly model users' news-specific behavioral heterogeneity, nor does it incorporate the temporal, structural, and comment-semantic signals that are crucial for characterizing how a collective forms around the target news item. In contrast, LeHT adaptively aggregates node and hyperedge information according to learned attention weights. Moreover, its behavior-aware node-to-node attention explicitly models behavioral interaction among users and integrates temporal participation, structural involvement, and comment-semantic responses into news-specific user representations. These heterogeneous user representations are then adaptively aggregated through node-to-hyperedge attention to form collective propagation behavior pattern representations. Therefore, the performance gap between LMCE+HGCN and LeHT indicates that the final gains of LeHT come not only from hyperedge-based aggregation itself, but also from its stronger ability to model collective propagation behavior patterns.

5.4.2. Effectiveness of context enhancement design

To further examine the contribution of different context enhancement designs, we construct the following variants of LeHT:

- **LeHT w/o CCE:** We remove the coarse-grained context enhancement (CCE) and retain only the fine-grained context enhancement (FCE).
- **LeHT w/o FCE:** We remove the fine-grained context enhancement (FCE) and retain only the coarse-grained context enhancement (CCE).
- **LeHT w/ pooling CE:** We replace the cross-attention-based context enhancement module with a naive pooling strategy that directly aggregates the original news representation and all generated context candidates.

As reported in Table 3, removing either CCE or FCE leads to performance drops on three datasets, indicating that context information at different granularities is important for understanding news semantics. Specifically, CCE provides background information about the related event, while FCE supplies descriptive information about the involved entities. These two types of context are complementary and jointly contribute to richer semantic understanding, which can be beneficial for fake news detection.

More importantly, LeHT w/ pooling CE performs worse than the full LeHT. This result shows that the benefit of context enhancement does not come simply from injecting more contextual information into the model. Instead, the cross-attention-based context enhancement module plays an essential role in selectively highlighting informative contextual candidates while suppressing irrelevant, noisy, or potentially hallucinated ones. In contrast, the pooling-based variant treats all generated context candidates more uniformly and cannot effectively distinguish truly useful contextual cues from less relevant ones. Therefore, the superior performance of LeHT further demonstrates the necessity of cross-attention-based context enhancement for robust and effective semantic enrichment.

5.5. Sensitivity analysis (RQ4)

5.5.1. Effect of LLM choice on LeHT

To examine the sensitivity of LeHT to the choice of LLM used for context generation, we construct the following variants:

- **LeHT w/ GPT-3.5:** We prompt GPT-3.5 to generate multi-granularity contexts in the LMCE module. GPT-3.5 is a closed-source large language model developed by OpenAI.
- **LeHT w/ GPT-4.0:** We prompt GPT-4.0 to generate multi-granularity contexts in the LMCE module. GPT-4.0 is the upgraded version of GPT-3.5.
- **LeHT w/ LLaMA-3:** We prompt LLaMA-3 to generate multi-granularity contexts in the LMCE module. Llama-3 is an open-source large language model proposed by Meta. In the experiment, we selected the 8B-parameter version and deployed it on local GPUs.

For a fair comparison, we use the same prompt template for all LLMs. As shown in Fig. 5, LeHT w/ GPT-4.0 achieves the best overall performance, indicating that stronger LLMs can provide higher-quality contexts and further improve detection accuracy. Meanwhile, LeHT w/ GPT-3.5 and LeHT w/ LLaMA-3 also outperform the comparative methods, suggesting that LeHT is not overly sensitive to the specific LLM choice. Even when using a relatively smaller LLM such as LLaMA-3, LeHT still outperforms the comparative methods. This suggests that the gains of LeHT are not primarily attributable to the use of a particularly strong LLM, but are also rooted in the proposed collective propagation behavior pattern modeling framework.

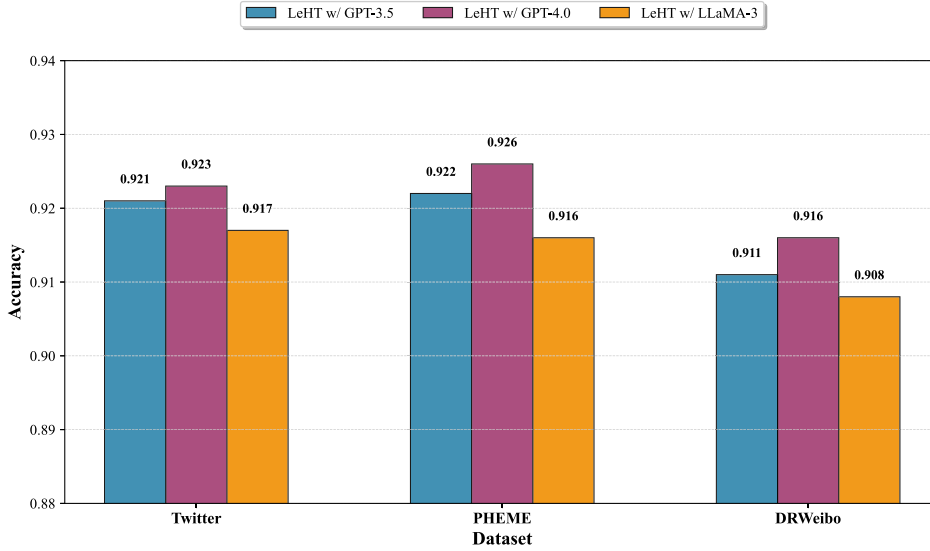


Fig. 5. The effect of different LLMs chosen on model performance.

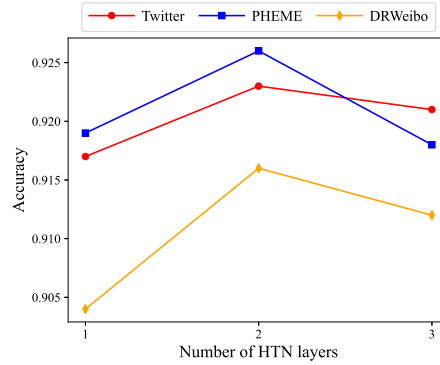


Fig. 6. The effect of different numbers of HTN layers on model performance.

5.5.2. Effect of the number of HTN layers on LeHT

In this subsection, we investigate how the number of HTN layers affects the performance of LeHT. Specifically, we vary the HTN layers from 1 to 3 and report the corresponding ACC on the three datasets. The results are shown in Fig. 6.

As can be observed, LeHT achieves strong performance when using two HTN layers, which yields the best overall results on three datasets. When the number of HTN layers increases to three, the performance decreases on all datasets. A plausible explanation is that the deeper HTN introduces more parameters and model capacity, which may lead to overfitting under the given training setting. Overall, these results suggest that a two-layer HTN provides the most favorable trade-off and is adopted as the default configuration in our experiments.

5.6. Time efficiency (RQ5)

To assess whether the performance gains of LeHT introduce significant time overhead, we evaluate the inference time of LeHT and its distilled variant, LeHT-D, and compare them with several representative state-of-the-art methods, including ARG, AD-GSCL, FGDGNN, and HG-SL, on the three datasets. Notably, to better reflect real-world scenarios, the inference time we measured includes both the LLM response time and the model forward propagation time. Table 4 reports the average inference time.

We can draw the following conclusions from the experiment results: 1. We found that LLM-assisted methods (including ARG and LeHT) require more inference time. This is because they need to query the LLM during inference and wait for its response. The response time of the LLM largely depends on external factors (e.g., service latency and network conditions). As LLMs become more efficient and widely deployed, response times may decrease. 2. LeHT exhibits slightly longer inference times than ARG but demonstrates significantly superior fake news detection performance. This validates the effectiveness and time efficiency of our

Table 4
Comparison of inference time for different models.

Method	Twitter	PHEME	DRWeibo
LeHT	3.24 s	3.19 s	3.27 s
LeHT-D	1.23 s	1.17 s	1.29 s
ARG	3.08 s	3.04 s	3.17 s
AD-GSCL	1.04 s	1.03 s	1.11 s
FGDGNN	1.13 s	1.07 s	1.14 s
HG-SL	1.21 s	1.12 s	1.23 s

proposed LeHT model. 3. The distilled version LeHT-D substantially reduces inference time while maintaining high fake news detection performance. This is attributed to knowledge distillation techniques transferring the LLM's knowledge enhancement capabilities to a lightweight model, significantly lowering computational costs without compromising performance. LeHT-D's inference time is comparable to other propagation-based methods (including AD-GSCL, FGDGNN, and HG-SL), yet its performance is markedly improved. This is because LeHT-D explicitly models higher-order correlations in collective propagation behavior through hypergraphs and introduces a hypergraph transformer network to capture highly discriminative representations of collective propagation patterns.

Overall, our proposed LeHT model achieves comparable performance gains to other LLM-assisted methods (such as ARG) without significantly increasing inference time. The distilled LeHT-D model notably reduces inference time while maintaining effective fake news detection capabilities. This greatly facilitates model deployment in cost-sensitive scenarios.

5.7. Visualization analysis (RQ6)

To provide an intuitive illustration of the effectiveness of collective-level propagation behavior patterns for fake news detection, we visualize the learned news representations using t-SNE (Maaten & Hinton, 2008). Specifically, we compare LeHT with two representative variants: LMCE, which relies on context-enhanced news semantics only, and LMCE+GTN, which uses a simple graph to model the propagation process and thus mainly captures individual-level propagation behaviors. We conduct the visualization on all three datasets, and the t-SNE results on Twitter, PHEME, and DRWeibo are shown in Fig. 7.

As can be observed, the representations learned by LeHT exhibit a clearer classification boundary of different labeled news items, with intraclass consistency and greater interclass separation, compared with its variants. This result suggests that collective-level propagation analysis can provide powerful clues for detecting fake news. In contrast, the embeddings produced by LMCE show more overlapping clusters and less distinct decision boundaries, indicating that news content semantics alone may be insufficient to distinguish carefully crafted fake news. Moreover, LMCE+GTN has more blurred classification boundaries than LeHT, implying that focusing solely on individual propagation behaviors while neglecting collective propagation behaviors is insufficient for fake news detection tasks.

5.8. Case study (RQ7)

To qualitatively verify that large language models (LLMs) can generate useful contextual information for understanding short news posts, we select three representative fake news items from Twitter, PHEME, and DRWeibo, and prompt the LLM to produce multi-granularity contexts. The generated contexts are reported in Table 5.

Overall, the LLM provides informative contexts that supply missing background information and clarify key entities, which can facilitate semantic understanding and help identify the news semantics that may trigger collective responses.

Case 1 (Twitter). The post claims that former U.S. President Barack Obama filed federal charges against George Zimmerman following the Trayvon Martin incident. The coarse-grained context summarizes the timeline of the incident and the subsequent acquittal, while the fine-grained context clarifies the roles of core entities (Obama, Trayvon Martin, and Zimmerman). The generated context notes that filing federal criminal charges is typically handled by the U.S. Department of Justice rather than by the President, which contradicts the claim in the post and provides a strong cue that the statement is likely false. This post may exploit existing social tensions (e.g., race and justice) to elicit stronger emotional reactions and engagement, which can facilitate rapid diffusion.

Case 2 (PHEME). The post alleges that football player Michael Essien contracted Ebola and that AC Milan confirmed it. The coarse-grained context provides background on the Ebola outbreak and mentions denials from both the individual and the club. These details directly conflict with the post's claim and therefore serve as informative evidence for identifying the rumor. Such content is also likely to attract public attention and trigger rapid diffusion due to its sensational nature (a well-known athlete and a severe epidemic).

Case 3 (DRWeibo). The post states that the girl assaulted in the Tangshan restaurant incident had died. The generated context summarizes the incident and emphasizes that official reports indicated the victim was in stable condition and out of danger, which is inconsistent with the post's assertion. This contradiction offers a clear cue for detection. Meanwhile, the post leverages emotionally charged language, which may amplify engagement and contribute to widespread dissemination.

In summary, these examples suggest that LLM-generated multi-granularity contexts can effectively compensate for missing background information in short posts and surface key inconsistencies and semantic cues, thereby supporting downstream fake-news detection.

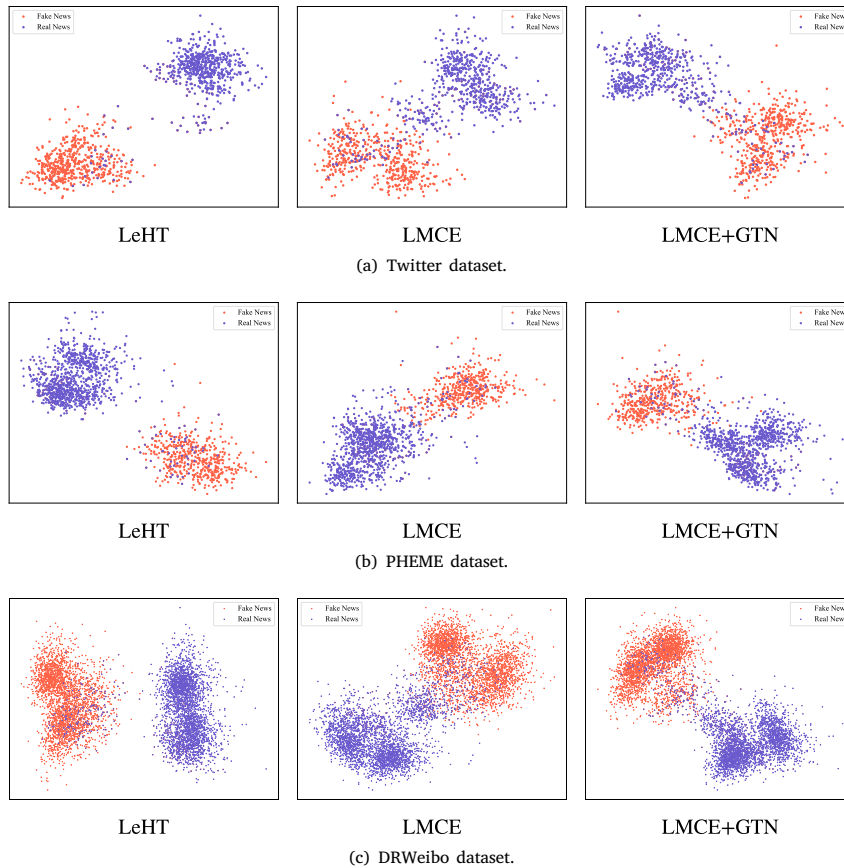


Fig. 7. T-SNE visualizations of the features learned by LeHT, LMCE, and LMCE+GTN on three datasets. Dots with the same color are within the same label. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

6. Discussion

6.1. Theoretical implications

This study offers important theoretical contributions to fake news detection research, as summarized in the following key points:

(1). We introduce collective-level propagation analysis as a new problem-solving perspective for fake news detection. Unlike existing studies that mainly detect fake news by mining textual semantics or individual-level propagation clues, this work emphasizes that fake news detection can also be addressed by analyzing collective propagation behavior patterns. This perspective is grounded in theories of information diffusion and social psychology. Prior studies suggest that news diffusion on social platforms is not merely a simple accumulation of isolated user actions; rather, it is shaped by collective propagation behaviors, in which users may make interdependent propagation decisions under the joint influence of news semantics and visible collective signals released by the already formed propagation collective. From this perspective, real and fake news differ not only in content but also in how collective propagation behavior patterns are formed and evolve. Therefore, this work extends the theoretical perspective of fake news detection from individual-level propagation analysis to collective-level propagation analysis and highlights the importance of collective propagation behavior patterns as a new source of discriminative evidence.

(2). We propose an LLM-enhanced hypergraph transformer framework to operationalize collective-level propagation analysis. To translate the above perspective into a concrete detection framework, we propose LeHT, which combines hypergraph modeling with a hypergraph transformer architecture for fake news detection. In contrast to conventional graph neural networks that mainly model pairwise interactions, LeHT leverages the flexible structure of hypergraphs to explicitly model collective propagation behaviors, where a single news item can connect multiple participating users. On this basis, the proposed hypergraph transformer network further captures higher-order dependencies through a three-level attention mechanism, including hyperedge-to-node attention, behavior-aware node-to-node attention, and node-to-hyperedge attention. In this way, LeHT provides a principled modeling framework for learning collective propagation behavior pattern representations, thereby extending the methodological foundation of propagation-based fake news detection from pairwise diffusion modeling to higher-order collective propagation modeling.

Table 5

Case study on multi-granularity contextual information generated by large language models.

Case 1. A fake news item from the Twitter dataset. Source Tweet Created in July 2013: News: <i>Obama files federal charges against George Zimmerman following acquittal in the shooting murder of Trayvon Martin.</i> Coarse-grained Contexts: - On February 26, 2012, a 17-year-old African American teenager, Trayvon Martin, was shot and killed by community patrol volunteer George Zimmerman during an altercation in Sanford, Florida. On July 13, 2013, a jury found Zimmerman not guilty. The case triggered intense debates about racial bias, gun rights, and judicial fairness. - Following the 2013 acquittal of George Zimmerman, Obama spoke about the Trayvon Martin case and racial issues in American society, calling for calm and reflection. However, this does not equate to Obama bringing federal charges against Zimmerman. Fine-grained Contexts: - Entity 1: Barack Obama, the 44th president of the United States. - Entity 2: Trayvon Martin, a 17-year-old high school student, African American. He was the victim of a shooting. - Entity 3: George Zimmerman, a 28-year-old white man of mixed Latin American descent. He was involved in the shooting incident. - Entity 4: Federal Charges, typically pursued by the U.S. Department of Justice (DOJ) and the Office of the U.S. Attorney. Formal proceedings involve indictments, criminal complaints, court filing information, etc.
Case 2. A fake news item from the PHEME dataset. Source Tweet Created in October 2014: News: <i>Breaking news: Ghana international AC Milan star Michael Essien contracted Ebola, the club confirmed.</i> Coarse-grained Contexts: - The 2014 Ebola outbreak in West Africa was the largest since the virus was first identified in 1976. The epidemic resulted in over 28,600 reported cases and approximately 11,325 deaths. - Michael Essien issued a public statement on Twitter confirming he is in good health and will resume training as scheduled the following day. - AC Milan issued a statement on its official website denying that Michael Essien had contracted the Ebola virus. Fine-grained Contexts: - Entity 1: Ebola, a rare but severe infectious disease that is often fatal. The 2014–2016 Ebola outbreak in West Africa was one of the largest Ebola epidemics in history. - Entity 2: Michael Essien, Ghanaian midfielder, played for AC Milan in October 2014. - Entity 3: Ghana, a country in West Africa. - Entity 4: AC Milan, an Italian football club.
Case 3. A fake news item from the DRWeibo dataset. Source Tweet Created in June 2022: News: <i>The girl who was beaten in Tangshan has died. This is heartbreaking news.</i> Coarse-grained Contexts: - On June 10, 2022, a violent assault occurred at a barbecue restaurant in Tangshan, Hebei Province, where multiple men beat and chased several women, drawing nationwide attention. - Police subsequently apprehended the individuals involved and handled the case according to the law. - Police report that the injured woman is in stable condition and out of danger. Fine-grained Contexts: - Entity 1: Tangshan City, the location where the incident occurred, is situated in Hebei Province, China. - Entity 2: The girl beaten, a victim of the assault incident at a Tangshan barbecue restaurant. She was beaten by multiple men at the restaurant.

(3). We leverage the rich world knowledge of LLMs to enrich news semantics. Unlike existing fake news detection methods that directly mine news semantics from brief news posts, we introduce LLMs as context providers to enhance news posts. Understanding the semantic triggers of collective responses is crucial for extracting collective-level propagation patterns, yet short news posts frequently lack the necessary contextual information. To mitigate semantic sparsity, we design an LLM-driven multi-granularity context enhancement module that leverages the broad world knowledge of LLMs to enrich news content. This deeper semantic understanding enables LeHT to better recognize content that is capable of eliciting collective reactions, thereby improving the capture of characteristic propagation patterns of fake news.

6.2. Practical implications

This study provides practical implications for misinformation governance on social platforms, as outlined below.

(1). LeHT can be applied in fake news detection systems in social platforms to support platform trustworthiness and information security. In practice, many platforms still rely heavily on manual fake news detection, which is time-consuming, labor-intensive, and struggles to handle massive data volumes. By capturing collective-level propagation behavior patterns, LeHT provides more effective detection performance. For cost-sensitive deployment, we further develop a distilled variant, LeHT-D, which removes the need for online LLM calls at inference time. This significantly reduces latency and operational costs while maintaining strong detection performance.

(2). LeHT can be applied in early intervention systems on social platforms to mitigate the harm caused by the widespread dissemination of fake news. Early intervention systems identify potentially harmful content during its initial dissemination phase and employ intervention actions to reduce its impact. Early detection experiments demonstrate that LeHT can verify the authenticity of fake news before its large-scale propagation, indicating its potential for application in early intervention systems.

(3). The LMCE module can be applied in content moderation systems on social platforms. Content moderation systems assess content veracity, especially when propagation signals are limited. However, social network news posts are brief and thus lack sufficient contextual information, which poses a challenge for content moderation. The LMCE module can prompt an LLM to generate semantically rich contextual information, thus mitigating semantic sparsity. Experiments show that the multi-granularity context

information generated by LLM is indeed important for understanding news semantics, which brings a bright prospect for LMCE to be applied to content auditing systems.

6.3. Future work

Future work can proceed along two main directions.

(1) Retrieval-augmented context enhancement. Although the current LMCE module improves semantic understanding by prompting an LLM to generate multi-granularity contextual candidates, LLM-generated context may still contain hallucinated or weakly grounded information. A promising direction is to incorporate retrieval-augmented generation (RAG) into the current framework. By retrieving relevant evidence from trustworthy external corpora, RAG can help ground contextual generation in more reliable knowledge, thereby reducing the risk of hallucination. Moreover, retrieval can provide the LLM with updated background information beyond its parametric memory, which is particularly important for fake news detection in rapidly evolving real-world environments. Integrating RAG with the current selective context enhancement framework may therefore further improve the reliability, factual grounding, and temporal adaptability of semantic enhancement.

(2) Dynamic hypergraph modeling of collective formation. The current LeHT models the collective propagation behaviors through hypergraphs. Although this design preserves higher-order collective dependencies and incorporates temporal behavior signals, it does not explicitly model how the collective structure itself evolves over time. An important future direction is to extend the current framework toward a dynamic hypergraph model, in which the collective formation process can be explicitly represented through time-evolving hypergraph structures. Such a formulation would make it possible to directly capture how users gradually join the propagation collective, how collective signals emerge and strengthen, and how collective propagation behavior patterns dynamically evolve during diffusion. This may further improve the ability of the model to characterize the formation process of collective behaviors in fake news propagation.

7. Conclusion

This study introduces a new perspective for fake news detection: collective-level propagation analysis, which is grounded in social psychology and information diffusion research. The proposed perspective aims to capture the collective propagation behavior patterns of news and to characterize the differences between real news and fake news in terms of collective propagation behavior patterns. Based on this perspective, we develop an LLM-enhanced hypergraph transformer framework (LeHT) that jointly exploits the strengths of large language models and hypergraph structures for collective-level analysis. Specifically, LeHT leverages the rich world knowledge of LLMs to enrich short social media posts with event- and entity-level context, facilitating a deeper understanding of news semantics that may trigger collective reactions. In addition, LeHT introduces a hypergraph transformer module to explicitly model higher-order correlations in collective propagation behaviors and to learn informative representations of collective propagation behavior patterns. Extensive experiments on three real-world datasets demonstrate the effectiveness of LeHT for fake news detection.

CRediT authorship contribution statement

Siqi Wei: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Formal analysis, Conceptualization. **Bin Wu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition. **Tao Zhou:** Writing – review & editing, Visualization, Software, Project administration, Formal analysis, Conceptualization. **Xuanning Liu:** Writing – review & editing, Visualization, Software, Formal analysis, Data curation. **Jiachen Tan:** Writing – review & editing, Visualization, Validation, Software, Methodology. **Di Liu:** Writing – review & editing, Software. **Guangyao Su:** Software.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

During the preparation of this work the authors used ChatGPT in order to improve language only. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Data availability

Data will be made available on request.

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